

Drought Triggers Training Document

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Drought Anticipatory Action System: Quick Start Guide

The Drought Anticipatory Action system operates in two modes:

1. **Hindcast Forecast Verification and Trigger Selection:** Used to verify forecast skill against historical observations and determine optimal trigger values for anticipatory action.
2. **Monthly Operational Forecasting:** Used for regular monthly forecasting once trigger values have been established.

1.1 Mode 1: Hindcast Forecast Verification & Trigger Selection

Run these commands in sequence to validate forecast quality and establish optimal trigger values:

```
# Step 1: Download required data
python 00-download-data.py --output-dir './' --keep-all-seas5
--only-current-month-seas5 4 --year 2025

# Step 2: Process SPI for SEAS5 forecasts and CHIRPS observations
python 01-run-process-spi.py --region-id kmj --mode both --use-local
--local-shapefile ../data/kmj_polygon.shp --chirps-file ../data
/chirps-v2.0.monthly.nc --seas51-main-file ../data
/3c58a474556eba4e1fd6a0d24e9824e8.grib --seas51-additional-files ../data
/47ed48882ef56a748a72999cle28baa4.grib --apply-mask --mask-buffer 0.25

# Step 3: Run forecast verification
python 02-run-fv-kmj.py

# Step 4: Generate bar plots
python 04-run-bar-plot.py

# Step 5: Create heatmap visualizations
python 05-run-heatmap.py

# Step 6: Plot SEAS51 forecasts
python 07-plot-sea51-forecast.py --region_id kmj --season JJA --lead_time 3 --year
2025 --month 4 --use_shpfile --shapefile_path '../data/kmj_polygon.shp'
--output_dir './'
```

1.2 Mode 2: Monthly Operational Forecasting

Once trigger values have been established, use the following steps for operational monthly forecasting:

```
# Step 1: Download required data
python 00-download-data.py --output-dir '.././data' --seas5-only
--only-current-month-seas5 3-5 --year 2025

# Step 2: Process SPI for SEAS5 forecasts only
# For May 2025
python 01-run-process-spi.py --region-id kmj --output-dir output --mode seas51
--use-local --obs-file kmj_obs_spi3.nc --local-shapefile .././data
/kmj_polygon.shp --seas51-main-file .././data
/3c58a474556eba4e1fd6a0d24e9824e8.grib --seas51-additional-files .././data
/seas5_precipitation_20250514_year2025_months_03_04_05.grib --apply-mask
--mask-buffer 0.25

# Step 3: Plot SEAS51 forecasts
python 07-plot-sea51-forecast.py --region_id kmj --season JJA --lead_time 3 --year
2025 --month 4 --use_shpfile --shapefile_path '.././data/kmj_polygon.shp'
--output_dir './'
# For May 2025
python 07-plot-sea51-forecast.py --region_id kmj --season JJA --lead_time 2 --year
2025 --month 5 --use_shpfile --shapefile_path '.././data/kmj_polygon.shp'
--output_dir './'

# Step 4: KMJ admin district level statistics
python 08-kmj-district-stats.py --input-netcdf
kmj_seas51_spi3_jja_eprob_2025_05.nc --admin-level admin2 --district-shapefile ..
/./data/Karamoja_9_districts.shp
```

1.3 Detailed Guide

1.3.1 Connecting to the Remote Server and Preparing the Environment

Before running any of the scripts, you need to connect to the remote server and set up your environment:

1. **Open a terminal application**

Open XMoTerm or another terminal application on your local machine.

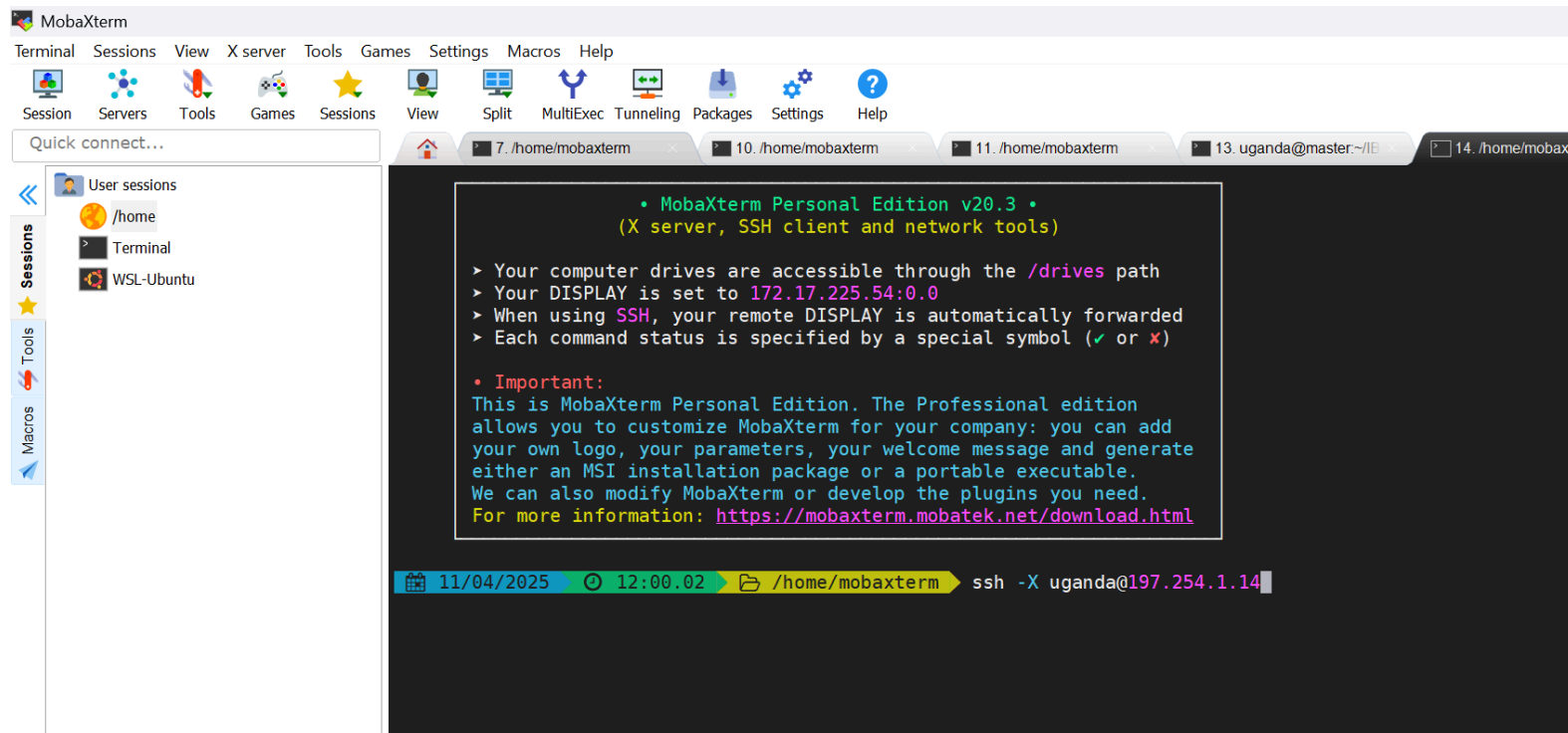


Figure 1.1. Opening XMoTerm on your local machine

2. Connect to the remote server with SSH

Use SSH to connect to the remote server with X forwarding enabled:

```
ssh -X uganda@197.254.1.14
```

You'll be prompted for a password.

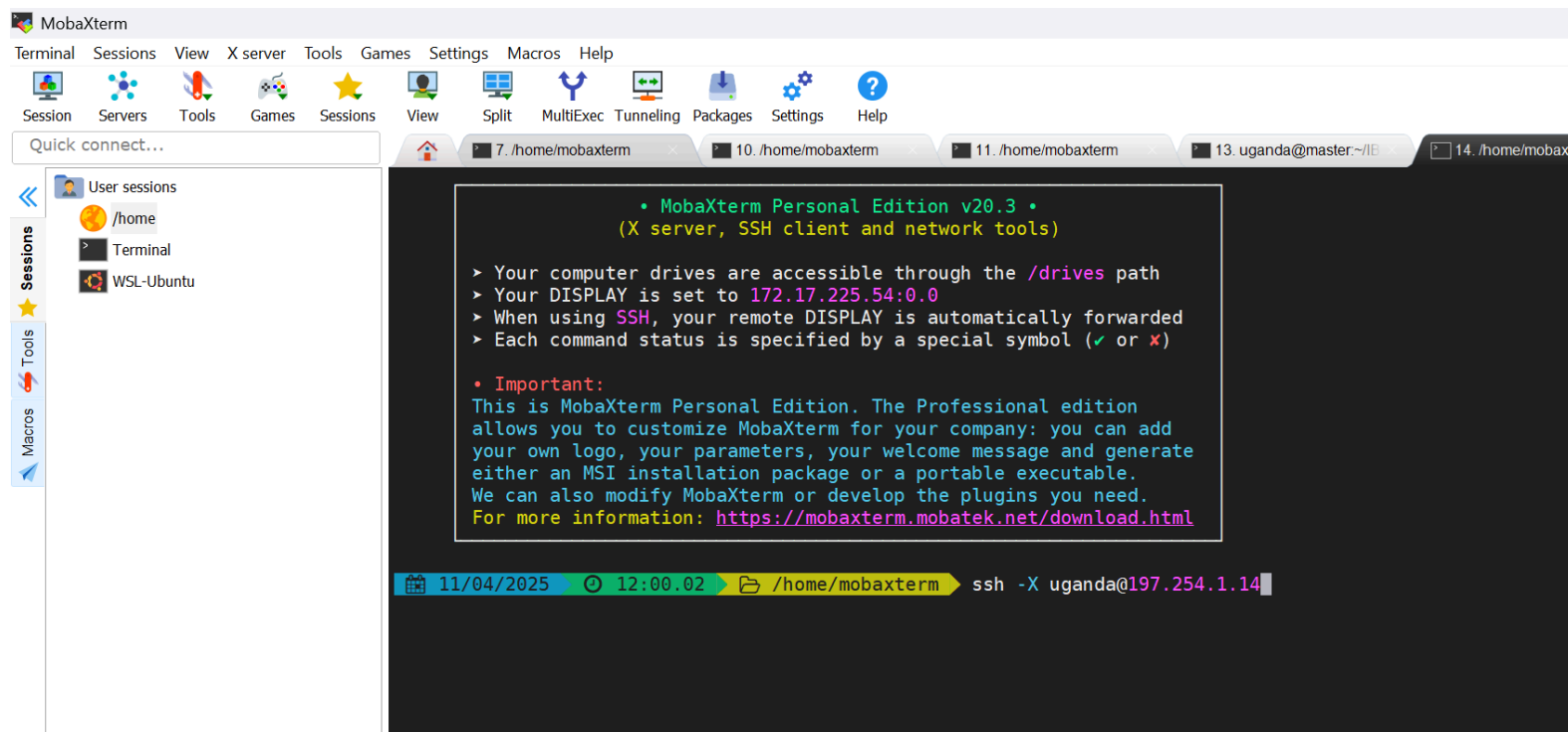


Figure 1.2. SSH login to the remote server

3. Activate the conda environment

Once connected, activate the specialized Python environment using micromamba:

```
micromamba activate drought_env
```

Your prompt should change to indicate the environment is active.

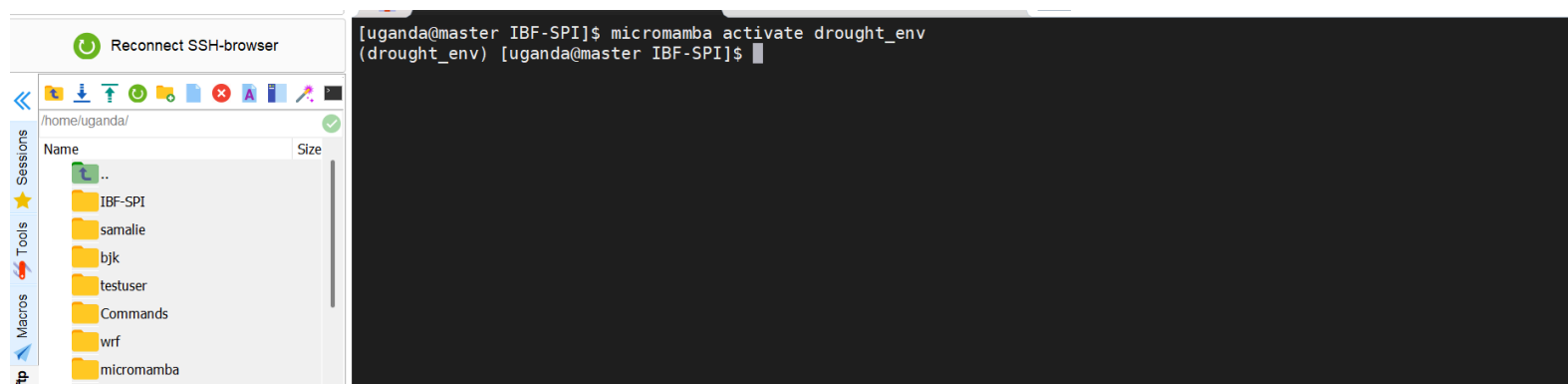


Figure 1.3. Activating the drought_env environment

4. Navigate to the working directory

Change to the directory containing the drought anticipatory action scripts:

```
cd /path/to/drought_scripts
```

Now you're ready to run the scripts in either mode.

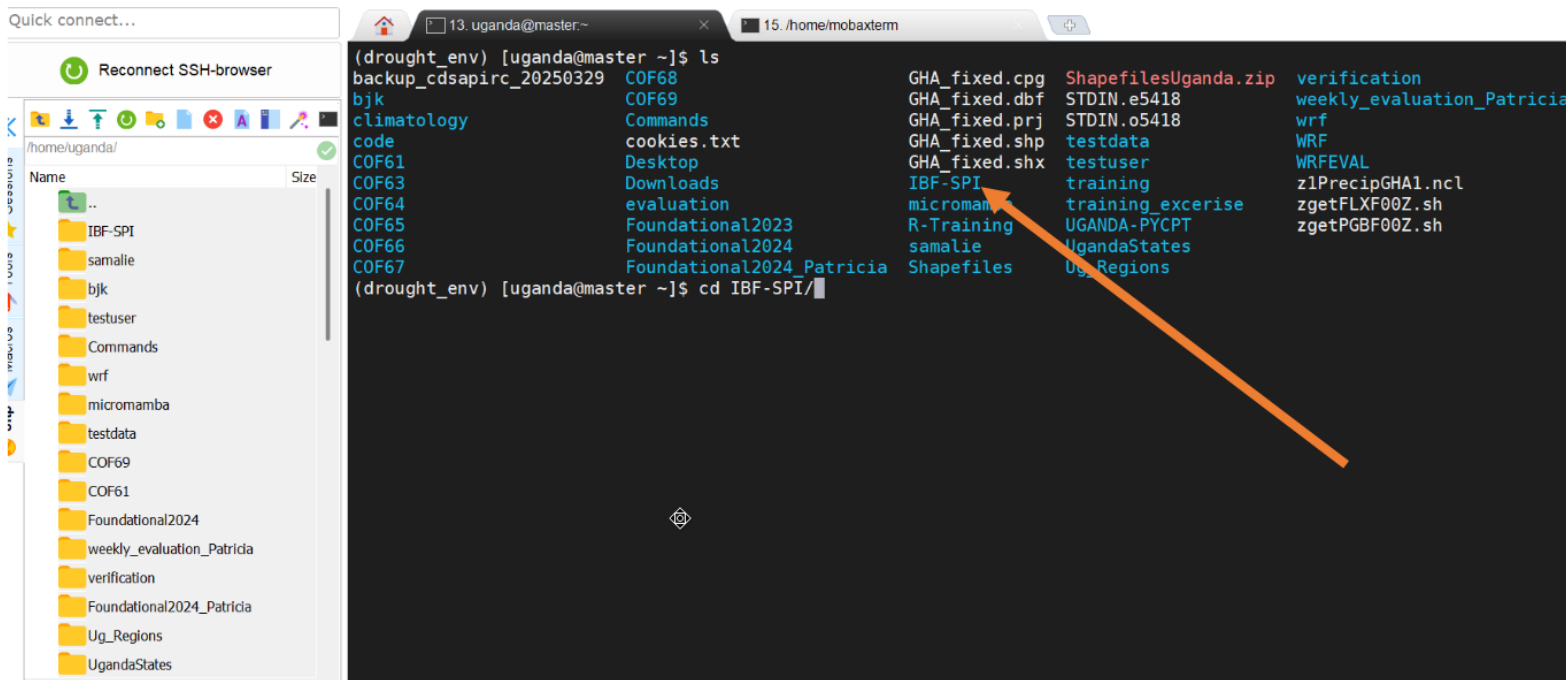


Figure 1.4. Navigating to the working directory

1.3.2 Step 1: Download ECMWF SEAS5 and CHIRPS Data

```
python 00-download-data.py --output-dir './' --keep-all-seas5
--only-current-month-seas5 4 --year 2025
```

Parameters explained:

- `--output-dir './'`: Specifies the directory where downloaded data will be saved
- `--keep-all-seas5`: Keeps all previously downloaded SEAS5 files instead of deleting older ones
- `--only-current-month-seas5 4`: Downloads SEAS5 data only for April (month 4)
- `--year 2025`: Specifies the year for the SEAS5 data
- `--seas5-only`: Download only SEAS5 data (no CHIRPS) - use for monthly operational mode

Additional options:

- `--chirps-only`: Download only CHIRPS data (no SEAS5)
- `--only-current-month-seas5 {1-12}`: Specify which month's data to download

Note This script automatically names files with date stamps (e.g., `seas5_precipitation_20250402.grib`) and CHIRPS data is only downloaded if it doesn't already exist.

1.3.3 Step 2: Process SPI (Standardized Precipitation Index)

For hindcast verification mode:

```
python 01-run-process-spi.py --region-id kmj --mode both --use-local
--local-shapefile ../data/kmj_polygon.shp --chirps-file ../data
```

```
/chirps-v2.0.monthly.nc --seas51-main-file ../data  
/3c58a474556eba4e1fd6a0d24e9824e8.grib --seas51-additional-files ../data  
/47ed48882ef56a748a72999c1e28baa4.grib --apply-mask --mask-buffer 0.25
```

For monthly operational forecasting:

```
python 01-run-process-spi.py --region-id kmj --mode forecast --use-local  
--local-shapefile ../data/kmj_polygon.shp --seas51-main-file ../data  
/3c58a474556eba4e1fd6a0d24e9824e8.grib --apply-mask --mask-buffer 0.25
```

Parameters explained:

- `--region-id kmj`: Specifies the region identifier (Karamoja in this case)
- `--mode both`: Processes both forecast and observation data (use for hindcast verification)
- `--mode forecast`: Processes only forecast data (use for monthly operational forecasting)
- `--use-local`: Uses local files instead of cloud storage
- `--local-shapefile ../data/kmj_polygon.shp`: Path to the shapefile defining the region
- `--chirps-file`: Path to CHIRPS observation data (only needed for hindcast verification)
- `--seas51-main-file`: Path to main SEAS51 forecast file
- `--seas51-additional-files`: Path to additional SEAS51 files (if needed)
- `--apply-mask`: Applies a spatial mask based on the shapefile
- `--mask-buffer 0.25`: Adds a 0.25-degree buffer around the masked region

Output: This script generates netCDF files with calculated SPI values.

Processing SPI values from forecast and observation data

1.3.4 Step 3: Run Forecast Verification (Hindcast Mode Only)

```
python 02-run-fv-kmj.py
```

This script runs the forecast verification process without requiring additional parameters. It:

- Calculates 2D and 1D forecast verification metrics
- Uses bootstrapping to calculate AUROC scores
- Generates CSV files containing verification statistics
- Subsets and saves trigger data for subsequent visualization

Note This step is only used in the hindcast verification mode, not in the monthly operational mode.

1.3.5 Step 4: Generate Bar Plots (Hindcast Mode Only)

```
python 04-run-bar-plot.py
```

This script creates bar plot visualizations comparing:

- Observed SPI values from CHIRPS data
- Forecast SPI values from SEAS5 data
- Decision tables with trigger performance metrics

Note This step is only used in the hindcast verification mode, not in the monthly operational mode.

1.3.6 Step 5: Create Heatmap Visualizations (Hindcast Mode Only)

```
python 05-run-heatmap.py
```

This script generates heatmap visualizations showing:

- Hit rates and false alarm ratios across different drought categories
- Trigger values for different lead times and regions
- Performance metrics for mild, moderate, and severe drought categories

Note This step is only used in the hindcast verification mode, not in the monthly operational mode.

1.3.7 Step 6/3: Plot SEAS51 Forecasts (Both Modes)

```
python 07-plot-sea51-forecast.py --region_id kmj --season JJA --lead_time 3 --year 2025 --month 4 --use_shpfile --shapefile_path '../data/kmj_polygon.shp' --output_dir './'
```

Parameters explained:

- `--region_id kmj`: Defines the region (Karamoja)
- `--season JJA`: Sets the season to analyze (June-July-August)
- `--lead_time 3`: Sets the forecast lead time in months
- `--year 2025` and `--month 4`: Sets the forecast initialization date
- `--use_shpfile`: Uses a shapefile to define the region boundaries
- `--shapefile_path`: Path to the shapefile
- `--output_dir './'`: Directory where output plots will be saved

Output: This script generates visualization maps showing ensemble forecasts, observations, and probabilistic forecasts for the selected region and season.

Note This step is used in both the hindcast verification mode and the monthly operational mode.

Mode 2: Monthly Operational Forecasting

Once trigger values have been established through the forecast verification process (Mode 1), the system transitions to monthly operational forecasting mode. This workflow is designed for regular monthly production of drought forecasts using pre-determined trigger values to support anticipatory action decisions.

This chapter details the operational workflow process, focusing on the scripts, their inputs, and outputs within the monthly forecast production cycle.

2.1 Overview

The operational forecasting workflow consists of four main steps:

1. **Data Acquisition:** Download the latest ECMWF SEAS5 data for specified months
2. **SPI Processing:** Process the forecast data to calculate SPI values
3. **Visualization:** Generate forecast maps and trigger exceedance visualizations
4. **District-Level Analysis:** Calculate district-level statistics for targeted decision-making

This streamlined workflow enables routine production of decision-relevant forecasts without repeating the extensive verification process each month.

2.2 Step 1: Download Required Data

The first step involves downloading the latest seasonal forecast data from ECMWF using the *00-download-data.py* script.

Script Overview:

The script connects to the Copernicus Climate Data Store (CDS) API to retrieve ECMWF SEAS5 seasonal forecast data for specific months and years. In operational mode, it focuses on downloading only the most recent initialization data needed for current forecasts, rather than the complete historical dataset required for verification.

Key Parameters:

- *-output-dir*: Directory where downloaded files will be saved
- *-seas5-only*: Flag to download only SEAS5 data (no CHIRPS observations)
- *-only-current-month-seas5*: Month(s) to download (3-4 means March-April)
- *-year*: Year for the forecast data (2025 in this example)

Command Example:

```
`bash python 00-download-data.py --output-dir '.././data' --seas5-only
--only-current-month-seas5 3-4 --year 2025 `
```

Output Files:

- SEAS5 GRIB file: *seas5_precipitation_YYYYMMDD_year2025_months_03_04.grib* - Includes total precipitation forecasts for specified months - Named with current date and requested month range - Stored in the specified output directory

Configuration Notes:

The script requires a configured CDS API key in the user's home directory (*~/cdsapirc*). The API configuration should contain:

```
`      url:          https://cds.climate.copernicus.eu/api/v2      key:
<your-user-id>:<your-api-key> verify: 0 `
```

SEAS5 data is organized by initialization months and includes forecasts for up to 6 months ahead. For operational forecasting, only the most recent initialization months are typically needed.

2.3 Step 2: Process SPI for SEAS5 Forecasts

The second step involves processing the downloaded SEAS5 data to calculate Standardized Precipitation Index (SPI) values using the *01-run-process-spi.py* script.

Script Overview:

This script processes SEAS5 forecast data to calculate SPI-3 values for a specific region. In operational mode (*--mode seas51*), it skips the processing of CHIRPS observation data and focuses only on the forecast component, using a pre-existing observation file for regridding purposes only.

Key Parameters:

- *--region-id*: Region identifier (e.g., 'kmj' for Karamoja)
- *--output-dir*: Directory for output files
- *--mode*: Set to 'seas51' for forecast-only operational mode
- *--use-local*: Flag to use local shapefile instead of cloud data
- *--obs-file*: Path to existing observation file for regridding reference
- *--local-shapefile*: Path to region boundary shapefile
- *--seas51-main-file*: Path to main SEAS5 GRIB file
- *--seas51-additional-files*: Additional SEAS5 files to include
- *--apply-mask*: Flag to apply geographic mask based on shapefile
- *--mask-buffer*: Buffer size in degrees around masked region (0.25°)

Note on Observation File Requirement:

Even in operational mode, an observation file (*--obs-file*) is still required as a reference for regridding the forecast data to the correct spatial resolution and grid alignment. This file should be taken from the Mode 1 verification process and remains static during operational forecasting - you don't need to update it each month. The script uses this file only to determine the target grid structure, not for verification purposes.

Command Example:

```
`bash python 01-run-process-spi.py --region-id kmj --output-dir output
--mode seas51 --use-local --obs-file kmj_obs_spi3.nc --local-shapefile ../
./data/kmj_polygon.shp --seas51-main-file .././data
/3c58a474556eba4e1fd6a0d24e9824e8.grib --seas51-additional-files .././data
/seas5_precipitation_20250514_year2025_months_03_04_05.grib --apply-mask
```

```
--mask-buffer 0.25 `
```

Output Files:

- Raw SPI NetCDF: *{region_id}_raw_seas51_spi3.nc* - Contains raw SPI-3 values for each ensemble member - Organized by initialization date and lead time
- RegridDED SPI NetCDF: *{region_id}_rgr_seas51_spi3.nc* - SPI-3 values regridDED to match observation grid spacing - Used for visualization and further analysis
- Masked SPI NetCDF: *{region_id}_rgr_seas51_spi3_masked.nc* - Final SPI-3 values with region mask applied - This is the primary output used for subsequent steps

Processing Steps:

1. Load SEAS5 forecast data from local GRIB files
2. Subset data to the region of interest
3. Calculate SPI-3 for each ensemble member and lead time
4. Regrid to match observation dataset resolution
5. Apply region mask using shapefile boundary
6. Save processed data to NetCDF files

2.4 Step 3: Plot SEAS51 Forecasts

The third step involves creating visualization maps of the forecast probabilities and trigger exceedance using the *07-plot-sea51-forecast.py* script.

Script Overview:

This script creates comprehensive visualization maps showing ensemble forecasts, empirical probabilities for different drought categories (moderate, severe, extreme), and binary trigger exceedance maps. It uses pre-determined trigger values (from Mode 1 verification) to indicate where forecasted probabilities exceed decision thresholds.

Key Parameters:

- *-region_id*: Region identifier (e.g., 'kmj')
- *-season*: Target season for analysis (e.g., 'JJA' for June-July-August)
- *-lead_time*: Forecast lead time in months
- *-year*: Forecast initialization year
- *-month*: Forecast initialization month
- *-use_shpfile*: Flag to use local shapefile for region boundaries
- *-shapefile_path*: Path to region boundary shapefile
- *-output_dir*: Directory for output visualizations

Command Examples:

```
For April 2025 initialization (3-month lead time to JJA): `bash python
07-plot-sea51-forecast.py --region_id kmj --season JJA --lead_time 3 --year
2025 --month 4 --use_shpfile --shapefile_path '../data/kmj_polygon.shp'
--output_dir './`
```

```
For May 2025 initialization (2-month lead time to JJA): `bash python
07-plot-sea51-forecast.py --region_id kmj --season JJA --lead_time 2 --year
2025 --month 5 --use_shpfile --shapefile_path '../data/kmj_polygon.shp'
--output_dir './`
```

Output Files:

- Empirical Probability NetCDF: *kmj_seas51_spi3_{season}_eprob_{year}_{month}.nc* - Contains ensemble probability forecasts for each drought category - Used for district-level analysis in next step
- Combined Visualization Map: *{region_id}_{season}_lt{lead_time}.png* - Multi-panel visualization showing:
 - Ensemble member SPI values (all 51 members)
 - Empirical drought probabilities for moderate, severe, and extreme categories
 - Binary trigger exceedance maps for each drought category
 - Colorbar legends and forecast metadata

Visualization Components:

1. **Ensemble Members:** Individual SPI maps for all ensemble members
2. **Empirical Probabilities:** - Percentage of ensemble members exceeding threshold for each category - Shown with classified color scales
3. **Trigger Maps:** - Binary maps (Yes/No) showing areas where probabilities exceed trigger values - Color-coded with green (below trigger) and red (exceeds trigger)
4. **Metadata:** - Initialization date, valid period, lead time - Trigger values used for each drought category - Region and dataset identification

2.5 Step 4: District-Level Statistics

The final step calculates district-level statistics to support targeted decision-making using the *08-kmj-district-stats.py* script.

Script Overview:

This script regrides forecasts to higher resolution (1km), overlays with administrative boundaries, and calculates district-level average probabilities. This enables decision-makers to identify specific districts at risk and allocate resources accordingly.

Key Parameters:

- *-input-netcdf*: Path to empirical probability NetCDF file from step 3
- *-admin-level*: Administrative level to use (e.g., 'admin2' for districts)
- *-district-shapefile*: Path to district boundary shapefile

Command Example:

```
`bash      python      08-kmj-district-stats.py      --input-netcdf
kmj_seas51_spi3_jja_eprob_2025_05.nc      --admin-level      admin2
--district-shapefile ../../data/Karamoja_9_districts.shp `
```

Output Files:

- District Statistics CSV: *{input_filename}_district_averages.csv* - Contains district-level average probabilities for each drought category - Formatted with one row per district - Values presented as percentages (0-100%)

Processing Steps:

1. Load empirical probability forecast from NetCDF
2. Regrid data to 1km resolution using bilinear interpolation

3. Load district boundaries from shapefile
4. Create spatial mask for each district
5. Calculate average probability values for each district and category
6. Format and save results to CSV file

Example Output Structure:

```
`      District,mod_prob,sev_prob,ext_prob      Abim,35.2,12.5,5.2
Amudat,42.1,18.7,7.3      Kaabong,28.4,9.8,3.5      Kotido,31.5,11.2,4.1
Moroto,45.3,21.4,9.2      Nabilatuk,38.7,15.6,6.4      Nakapiripirit,33.9,13.1,5.8
Napak,29.8,10.5,3.9 `
```

2.6 Operational Implementation

The operational forecasting workflow is typically executed monthly when new SEAS5 forecasts become available. The process follows these general guidelines:

1. **Timing:** - Run at the beginning of each month after new SEAS5 forecasts are published - Process multiple lead times to establish forecast consistency
2. **Decision Support Products:** - Map visualizations for technical audience and decision-makers - District-level statistics for targeted resource allocation - Binary trigger information for clear “act or don’t act” decisions
3. **Interpretation Guidelines:** - Compare forecasts from consecutive months for consistency - Consider forecasts in context of verified skill for that season/lead time - Focus on highest-skill months and lead times as determined in Mode 1
4. **Technical Requirements:** - Python environment with required libraries (xarray, xesmf, matplotlib, etc.) - Access to ECMWF’s CDS API (configured with valid credentials) - Administrative boundary shapefiles for target region - Pre-determined trigger values from verification process

By following this streamlined operational workflow, the system produces regular, consistent forecast products that support anticipatory action against droughts without requiring the extensive verification process to be repeated.

Anticipatory Action for Drought Early Warning

3.1 Concept of Anticipatory Action (AA)

Anticipatory Action (AA) represents a paradigm shift in humanitarian response, where actions are taken before a disaster occurs based on forecasts rather than after impacts materialize. In drought contexts, AA leverages the slow-onset nature of droughts to implement interventions during the critical “window of opportunity” between a reliable forecast and the onset of drought impacts.

According to Guimarães Nobre et al. (2023), AA enables the implementation of predefined actions triggered by forecasts that exceed specific thresholds, facilitating early risk reduction. This approach links seasonal forecasting information with predetermined contingency plans, actors, and funding—often referred to as Forecast-based Financing (FbF). Unlike traditional humanitarian responses that mobilize after a crisis, AA systems preposition resources and initiate interventions when forecast indicators reach trigger values, potentially reducing losses to livelihoods and infrastructure while providing more dignified support to communities.

3.2 Ensemble Prediction Systems

Ensemble Prediction Systems (EPS) form the backbone of modern seasonal forecasting, addressing inherent uncertainties in weather and climate prediction. These systems generate multiple forecasts (ensemble members) by varying initial conditions, model physics, or both. The European Centre for Medium-Range Weather Forecasts’ SEAS5 system, for instance, provides 25-51 ensemble members with lead times up to 7 months.

As described by Guimarães Nobre et al. (2024), ensemble forecasts allow for probabilistic interpretation of potential outcomes, crucial for risk-based decision-making. Each ensemble member represents an equally likely future scenario, collectively providing information about forecast uncertainty and the range of possible outcomes. This probabilistic approach is essential for AA systems where decisions involve significant resource allocation under uncertainty.

3.3 Empirical Probability Calculation

For drought monitoring and forecasting, empirical probability calculations convert ensemble forecasts into decision-relevant metrics. The process involves:

1. Calculating drought indicators (e.g., Standardized Precipitation Index) for each ensemble member
2. Determining what proportion of ensemble members exceed predefined thresholds
3. Deriving empirical probabilities that represent the likelihood of drought conditions

These empirical probabilities become the basis for triggers in AA systems. According to Guimarães Nobre et al. (2023), the empirical probability approach enables quantitative assessment of forecast uncertainty, allowing decision-makers to balance the risks of action versus inaction.

3.4 Role of Long-term Verification

Long-term verification forms the cornerstone of trust and justification for any AA system. By systematically comparing historical forecasts with observed outcomes over many years, verification:

1. Quantifies forecast quality through metrics like hit rates and false alarm ratios
2. Establishes appropriate trigger values based on performance statistics
3. Builds confidence among stakeholders and donors in the AA approach
4. Supports continuous improvement of the forecasting system

As emphasized in both papers, verification analysis provides the evidence base needed to justify anticipatory investments. It ensures that actions are triggered at appropriate probability thresholds that balance the risks of missed events against false alarms—a critical consideration when resources are limited and each intervention must be carefully justified.

3.5 AUROC: A Key Metric for Forecast Verification

The Area Under the Receiver Operating Characteristic curve (AUROC) represents a fundamental metric in the verification of probabilistic forecasts for Anticipatory Action (AA) systems. This metric addresses a central question: does the forecasting system effectively discriminate between drought and non-drought events?

3.5.1 AUROC Concept and Calculation

The AUROC score quantifies a forecast's ability to discriminate between binary outcomes (drought /no drought) across different probability thresholds. As described by Guimarães Nobre et al. (2023), the calculation involves setting a range of probability thresholds for converting forecasts into binary predictions, then evaluating how well these predictions match observations.

The resulting curve plots the hit rate (proportion of observed droughts correctly forecast) against the false alarm rate (proportion of non-droughts incorrectly forecast as droughts) at various threshold levels. The area under this curve—the AUROC score—ranges from 0 to 1, where:

- 0.5 indicates no skill (equivalent to random guessing)
- 1.0 indicates perfect discrimination
- Values below 0.5 suggest systematically incorrect forecasts

3.6 Quality Threshold for Anticipatory Action

In operational AA systems for drought, an AUROC score of 0.5 serves as a critical quality threshold. As implemented in the “Ready, Set & Go!” system (Guimarães Nobre et al., 2024), forecasts with AUROC scores below 0.5 are considered unsuitable for triggering anticipatory actions, as they provide no added value over random chance.

This threshold represents a minimum standard for forecast quality—one that ensures actions are based on information with at least some predictive skill. When evaluating potential triggers for AA, only those derived from forecasts with AUROC > 0.5 are considered, creating a first-level quality control in the system design.

3.7 Bootstrapping for Uncertainty Assessment

To account for sampling uncertainty in AUROC estimation, the verification process employs bootstrapping techniques. By repeatedly resampling the forecast-observation pairs with replacement (typically 1,000 iterations), a distribution of AUROC scores is generated, allowing for:

1. Estimation of confidence intervals around the AUROC score
2. Assessment of statistical significance of skill above the 0.5 threshold
3. More robust decision-making about forecast quality

This approach acknowledges that verification statistics based on limited historical records contain inherent uncertainty, particularly for rare events like severe droughts.

3.8 Role in Trigger Selection

Beyond serving as a quality filter, AUROC scores influence the selection of optimal trigger values. Higher AUROC scores indicate greater potential for skillful discrimination, allowing for more confident selection of triggers based on other performance metrics like hit rates and false alarm ratios.

In the context of long-term verification, AUROC provides a foundation for justifying AA initiatives to stakeholders and funders by demonstrating that the underlying forecasting system possesses genuine predictive skill, making it a suitable basis for humanitarian decision-making.

3.9 References

1. Nobre, Gabriela Guimarães, et al. "Forecasting, thresholds, and triggers: towards developing a forecast-based financing system for droughts in Mozambique." *Climate Services* 30 (2023): 100344.
2. Guimarães Nobre, Gabriela, et al. "Ready, Set & Go! An anticipatory action system against droughts." *Natural Hazards and Earth System Sciences* 24.12 (2024): 4661-4682.

Essential Tools

This document provides a brief introduction to the fundamental tools needed for developing Standardized Precipitation Index (SPI) forecasting systems and drought monitoring triggers: Python programming language, Git version control with GitHub, and Jupyter Notebooks for interactive analysis.

4.1 Introduction to Python

Python has become the programming language of choice for scientific computing and data analysis due to its versatility, readability, and extensive ecosystem of libraries.

Key Concepts:

- **Scientific Libraries:** Python offers powerful packages like NumPy, SciPy, and pandas for processing and analyzing large datasets. Libraries such as xarray excel at handling multidimensional labeled data common in environmental sciences.
- **Statistical Processing:** A comprehensive suite of statistical libraries enables everything from basic descriptive statistics to complex machine learning models and time series analysis.
- **Geospatial Analysis:** Python's well-developed GIS libraries (GeoPandas, Shapely, Rasterio) support sophisticated spatial operations, coordinate transformations, and integration with remote sensing data.

Python 3.8 or newer is recommended for scientific applications, as recent versions offer performance improvements and newer features. The Anaconda distribution (<https://www.anaconda.com/products/distribution>) provides a convenient way to install Python along with most commonly used scientific libraries in a single package.

4.2 Git and GitHub for Scientific Reproducibility

Version control is fundamental to scientific computing, providing the infrastructure for reproducible research, rigorous verification, and transparent collaboration.

Key Concepts:

- **Code Versioning:** Git creates a detailed history of all changes, enabling scientists to track the evolution of analysis methods and maintain provenance of results.
- **Reproducibility:** By capturing both code and configuration files at each stage, researchers can precisely recreate analytical environments and verify that results are consistent across different systems and time periods.
- **Scientific Collaboration:** GitHub facilitates peer review of scientific code, enables geographically distributed teams to work on shared datasets, and supports the open science movement through

public repositories.

- **Verification Workflows:** Through branching and pull requests, teams can implement rigorous verification protocols where new analytical methods or triggers must be independently tested before being incorporated into operational systems.

In fields where computational results inform policy decisions or emergency response, version control provides the audit trail necessary for scientific credibility and accountability. Git's ability to document exactly how and when analytical thresholds were developed makes it particularly valuable for creating verified trigger systems in early warning applications.

The current code base along with the source files to create the documentaiton in pdf is stored in github link (<https://github.com/icpac-igad/ibf-thresholds-triggers/tree/kmj>) where the kmj is the current branch name.

4.3 Jupyter Notebooks as a Python Learning Tool

Jupyter Notebooks provide an excellent environment for learning Python and developing data analysis skills through interactive computing.

Key Concepts:

- **Interactive Learning:** Notebooks enable immediate execution of code cells, allowing learners to experiment with Python concepts and see results instantly.
- **Code and Documentation Together:** The ability to combine executable code with explanatory text makes notebooks ideal for self-guided learning and tutorials.
- **Visualization Integration:** Built-in support for charts and graphs allows beginners to easily visualize their data without switching between different tools.
- **Incremental Development:** The cell-based structure encourages breaking problems into manageable steps, helping new programmers develop good coding practices.

Jupyter Notebooks lower the barrier to entry for Python beginners by providing a more intuitive interface than traditional IDEs. The immediate feedback loop accelerates the learning process and helps users build confidence in their programming skills.

To get started with Jupyter, install it via pip or conda:

```
pip install notebook
```

Then launch it from your terminal:

```
jupyter notebook
```

4.4 Python Scripts and Jupyter Notebooks

Python scripts form the backbone of data analysis workflows, while Jupyter Notebooks provide an interactive environment to develop and test these scripts.

Python Scripts:

- **Structure:** Python scripts are plain text files with a `.py` extension containing Python code that can be executed from start to finish.
- **Execution:** Scripts can be run from the command line using the Python interpreter:

```
python script_name.py
```

- **Modularity:** Well-designed scripts can be imported and reused across different projects,

promoting code organization and efficiency.

- **Automation:** Scripts can be scheduled to run automatically or integrated into data processing pipelines.

Jupyter Notebooks:

- **Script Development:** Notebooks excel as a development environment where you can write and test code before converting it to production scripts.
- **Cell-Based Execution:** Unlike scripts that run from top to bottom, notebooks allow executing individual cells in any order, helping identify and fix issues.
- **Converting Between Formats:** Notebooks can be converted to Python scripts using the `nbconvert` tool:

```
jupyter nbconvert --to python notebook.ipynb
```

- **Importing Scripts:** Python scripts can be imported into notebooks using standard import statements, enabling you to test functions from your scripts interactively.

4.5 Integration for Operational Drought Forecasting

These three technologies work together to create robust drought forecasting systems:

1. **Python** handles the core calculations for SPI forecasting, statistical verification, and trigger development
2. **Git/GitHub** ensures forecast model integrity, enables collaboration, and tracks methodological changes
3. **Jupyter Notebooks** facilitates exploratory analysis, verification testing, and creation of drought forecast bulletins

By mastering these complementary tools, meteorologists and hydrologists can develop more reliable, transparent, and actionable drought early warning systems that connect seasonal forecasts to humanitarian decision-making.

Overview of Python Libraries for Forecast Verification

5.1 Core Libraries for Forecast Analysis

5.1.1 xarray

Essential for handling multidimensional arrays with labeled coordinates. Provides powerful data structures for working with forecast and observational gridded data across multiple time periods and ensemble members.

5.1.2 xclim

Specialized for climate indices calculation, particularly the `standardized_precipitation_index` function, which is fundamental for translating precipitation data into drought indicators using gamma distribution fitting.

5.1.3 xskillscore

Provides verification metrics specific to Earth science data. Used for calculating ROC curves and skill scores when evaluating forecast performance against observations.

5.1.4 xhistogram

Enables efficient histogram generation along multiple dimensions for gridded data. Critical for categorizing forecast and observation pairs into contingency tables for binary verification.

5.1.5 xesmf

Handles regridding between different spatial resolutions. Essential for aligning ECMWF SEAS5 forecast data with CHIRPS observational data before verification can begin.

5.1.6 xbootstrap

Implements bootstrapping methods for uncertainty quantification. Used to generate confidence intervals for AUROC scores and other verification metrics.

5.2 Statistical and Geospatial Processing

5.2.1 numpy

Provides the mathematical foundation for array operations. Handles the numerical calculations underpinning all verification metrics and statistical analyses.

5.2.2 pandas

Manages tabular data structures for storing verification results. Facilitates filtering, aggregating, and organizing trigger statistics by threshold level and lead time.

5.2.3 dask

Enables parallel computing and lazy evaluation for large datasets. Critical for handling the memory-intensive forecast ensemble calculations across multiple lead times.

5.2.4 regionmask

Creates masked arrays based on geographic boundaries. Essential for focusing the analysis on specific regions like Karamoja while excluding irrelevant grid cells.

5.2.5 geopandas

Extends pandas to work with geospatial data. Handles the shapefile operations needed to define region boundaries and create regional masks.

5.2.6 shapely

Provides computational geometry operations. Used for manipulating region polygons, calculating bounds, and applying buffers around analysis regions.

5.3 Visualization

5.3.1 matplotlib

Creates the foundational plots and visualizations. Generates time series, heatmaps, and spatial maps of SPI values and forecast skill.

5.3.2 seaborn

Builds on matplotlib to create more attractive statistical visualizations. Particularly useful for the heatmaps showing hit rates and false alarm ratios.

5.3.3 altair

Provides a declarative visualization syntax. Used for creating interactive and linked visualization components showing forecast performance.

5.3.4 vl_convert

Converts Vega-Lite specifications (from Altair) to static image formats like PNG. Essential for including visualizations in reports.

5.4 Cloud Data Access

5.4.1 google.oauth2

Manages authentication for accessing cloud-stored data. Enables secure access to forecast and observation datasets stored in Google Cloud Storage.

5.4.2 fsspec

Provides a unified file system interface. Allows consistent access to data regardless of whether it's stored locally or in cloud storage.

These libraries work together to form a comprehensive toolkit for drought forecast verification, enabling the entire workflow from data retrieval through processing, verification, and visualization of results.

Setting Up MicroMamba and Accessing Jupyter Notebook Remotely

This guide will walk you through setting up MicroMamba and accessing Jupyter Notebook from a remote computer using MobaXterm, a popular terminal client for Windows users.

6.1 Part 1: Setting Up MicroMamba

6.1.1 Installing MicroMamba

Linux/macOS

1. Open your terminal
2. Run the following command to install MicroMamba:

```
"${SHELL}" <(curl -L micro.mamba.pm/install.sh)
```

3. Follow the on-screen prompts to complete the installation
4. Close and reopen your terminal, or run `source ~/.bashrc` to apply the changes

Windows

MicroMamba also has Windows support. For Windows, we recommend using PowerShell:

1. Open PowerShell
2. Run the following commands:

```
Invoke-WebRequest -URI https://micro.mamba.pm/api/micromamba/win-64/latest
-OutFile micromamba.tar.bz2
tar xf micromamba.tar.bz2

MOVE -Force Library\bin\micromamba.exe micromamba.exe
.\micromamba.exe --help

# You can use e.g. $HOME\micromambaenv as your base prefix
$Env:MAMBA_ROOT_PREFIX="C:\Your\Root\Prefix"

# Invoke the hook
.\micromamba.exe shell hook -s powershell | Out-String | Invoke-Expression

# ... or initialize the shell
.\micromamba.exe shell init -s powershell -r C:\Your\Root\Prefix
```

6.1.2 Creating an Environment

1. If you have an environment file (environment.yml):

```
micromamba create -f environment.yml
```

2. If you don't have an environment file, you can create one with specific packages:

```
micromamba create -n drought_env python==3.12
```

6.1.3 Activating the Environment

```
micromamba activate drought_env
```

6.1.4 Installing Additional Packages

1. Install packages from conda-forge channel:

```
micromamba install netcdf4 -c conda-forge
```

2. Install packages using pip:

```
pip install xclim==0.56.0
```

6.2 Part 2: Remote SSH Access - For Windows and Linux Users

6.2.1 For Windows Users: Using MobaXterm

MobaXterm is a powerful terminal emulator for Windows that includes SSH, SFTP, and X11 forwarding capabilities.

1. **Install MobaXterm:**

- Download from [MobaXterm website](#)
- Choose either Installer edition (recommended) or Portable edition
- Follow the installation wizard if using the Installer edition

2. **Create an SSH Connection:**

- Launch MobaXterm
- Click on "Session" in the top-left corner
- Select "SSH" from the available session types
- Enter your remote server details: - Remote host: your_remote_server_ip - Username: your_username - Port: 22 (default)
- Optionally configure SSH key authentication in the "Advanced SSH settings" tab
- Save your session for future use

3. **Set Up Port Forwarding for Jupyter:**

- While creating your SSH session, click on the "Tunneling" tab
- Click "Add a new port forwarding"
- Configure: - Forward port: 4888 (local machine) - Remote server: localhost - Remote port: 4888 (remote machine)

- Save these settings with your session

4. Connect to Your Server:

- Click on your saved session in the left sidebar
- Enter your password when prompted (if not using key authentication)
- You now have terminal access to your remote server

5. Additional MobaXterm Features:

- Built-in SFTP browser in the left panel for easy file transfers
- Multiple tabs for working with several connections
- Saved sessions for quick access to your servers

6.2.2 For Linux Users: Using Terminal

Linux systems come with built-in terminal and SSH capabilities.

1. Open Terminal:

- Use your system's application launcher or press Ctrl+Alt+T

2. Connect to Remote Server:

- Use the SSH command:

```
ssh username@remote_machine_ip
```

- Enter your password when prompted

3. Set Up SSH Keys for Passwordless Login (optional but recommended):

- Generate an SSH key pair (if you don't already have one):

```
ssh-keygen -t rsa -b 4096
```

- Copy your public key to the remote server:

```
ssh-copy-id username@remote_machine_ip
```

- Now you can connect without entering a password

4. Set Up Port Forwarding for Jupyter:

- When connecting, add the port forwarding parameter:

```
ssh -L 4888:localhost:4888 username@remote_machine_ip
```

- This creates a secure tunnel from your local port 4888 to the remote server's port 4888

5. Using Terminal Multiplexers (recommended):

- Install and use tmux or screen for session persistence:

```
# Install tmux
sudo apt-get install tmux # For Debian/Ubuntu
# or
sudo yum install tmux     # For RedHat/CentOS

# Start a tmux session
tmux

# Detach from session (keeps running in background)
```

```
# Press Ctrl+b then d
# Reattach to session
tmux attach
```

Both Windows and Linux users will follow the same remaining steps for activating the MicroMamba environment and starting Jupyter Notebook as described in Part 3. Please refer to the attached “Using MobaXterm for Remote Access and Jupyter Notebook” guide for detailed instructions with visual references.

6.3 Part 3: Setting Up Jupyter Notebook for Remote Access

6.3.1 Starting Jupyter Lab on the Remote Machine

1. Connect to your remote machine using MobaXterm as described in the attached guide
2. Activate your environment:

```
micromamba activate drought_env
```

3. Start Jupyter Lab specifying a port (e.g., 4888):

```
jupyter lab --no-browser --port=4888
```

4. You'll see output containing a URL with a token. It will look something like:

```
http://localhost:4888/?token=abcdef123456...
```

Note this URL for the next step.

6.3.2 Accessing Jupyter Lab from Your Local Browser

1. Open a web browser on your Windows computer
2. Paste the URL with the token from step 1:

```
http://localhost:4888/?token=abcdef123456...
```

or simply go to:

```
http://localhost:4888
```

and enter the token when prompted.

6.4 Troubleshooting Tips

6.4.1 Port Already in Use

If port 4888 is already in use, choose a different port:

```
jupyter lab --no-browser --port=4889
```

Remember to adjust your port forwarding in MobaXterm accordingly.

6.4.2 Connection Issues

- Make sure your firewall allows connections to the specified port

- Verify that the Jupyter server is actually running
- Check if another Jupyter instance is already using that port

6.4.3 Session Disconnects

Configure SSH keepalive in MobaXterm:

- Go to Settings → SSH
- Set “SSH keepalive” to a value like 30 seconds

6.5 Setting Up a Password for Jupyter

For easier access in the future:

1. Set up a password:

`jupyter server password`

2. Enter and confirm your password
3. Start Jupyter as usual, and now you can use your password instead of the token

6.6 Using Your Remote Jupyter Environment

1. Your files and data on the remote machine will be accessible through the Jupyter interface
2. Any packages installed in your `drought_env` environment will be available in notebooks
3. Use MobaXterm’s SFTP browser (left sidebar) to easily upload/download files between your local and remote machines

For more detailed instructions on using MobaXterm for SSH connections and managing remote Jupyter sessions, please refer to the attached documentation.

Acquiring Data for Drought Forecasting: CHIRPS and ECMWF SEAS5

This guide explains how to download the two essential datasets required for drought forecasting using Standardized Precipitation Index (SPI):

1. **CHIRPS** - Climate Hazards Group InfraRed Precipitation with Station data (observational dataset)
2. **ECMWF SEAS5** - Seasonal forecast system from the European Centre for Medium-Range Weather Forecasts

7.1 Downloading CHIRPS Observational Data

CHIRPS version 3.0 provides quasi-global rainfall data from 1981 to near-present, making it ideal for drought monitoring.

Data Access Options:

1. **Direct Web Download**

CHIRPS data can be downloaded directly from the Climate Hazards Center website:

<https://data.chc.ucsb.edu/products/CHIRPS/v3.0/>

2. **Python Implementation with wget**

```
import os
import subprocess

def download_chirps(output_dir="./data", filename="chirps-v3.0.monthly.nc"):
    """
    Download CHIRPS precipitation data using wget

    Args:
        output_dir: Directory to save the downloaded data
        filename: Name for the output file

    Returns:
        str: Path to the downloaded file
    """
    print("Downloading CHIRPS precipitation data...")

    # Create output directory if it doesn't exist
    os.makedirs(output_dir, exist_ok=True)

    output_file = os.path.join(output_dir, filename)
    url = "https://data.chc.ucsb.edu/products/CHIRPS/v3.0/monthly/global/netcdf/chirps-v3.0.monthly.nc"
```

```

try:
    # Check if file already exists
    if os.path.exists(output_file):
        print(f"CHIRPS data already exists at: {output_file}")
        return output_file

    # Use wget to download the file
    subprocess.run(["wget", url, "-O", output_file], check=True)
    print(f"CHIRPS data downloaded successfully to: {output_file}")
    return output_file
except subprocess.CalledProcessError as e:
    print(f"Error downloading CHIRPS data: {e}")
    return None
except Exception as e:
    print(f"Unexpected error: {e}")
    return None

# Example: Download CHIRPS data
chirps_file = download_chirps()

# Open the data with xarray
import xarray as xr
ds = xr.open_dataset(chirps_file)
print(ds)

```

7.2 ECMWF SEAS5 Forecast Data from the Climate Data Store (CDS)

SEAS5 is ECMWF's fifth generation seasonal forecast system, providing global forecasts with lead times up to 7 months.

CDS Account Creation:

1. Register for a CDS Account

Visit the Copernicus Climate Data Store (CDS) website:

```
https://cds.climate.copernicus.eu/user/register
```

Complete the registration form and agree to the Terms and Conditions.

2. Generate an API Key

After registration:

- Log in to your CDS account
- Go to your profile page (click on your username in the top-right corner)
- Scroll down to the "API key" section
- The page will display your User ID and API Key

3. Configure the CDS API Client

Create a file called `.cdsapirc` in your home directory with the following contents:

```

$ cat ~/.cdsapirc
url: https://cds.climate.copernicus.eu/api/v2
key: <your-user-id>:<your-api-key>
verify: 0

```

Replace `<your-user-id>` and `<your-api-key>` with the values from your profile.

Python Implementation for ECMWF SEAS5 Download:

```

import os
import datetime
import cdsapi
import glob

def download_seas5(output_dir="./data", filename_prefix="seas5_precipitation_"):
    """
    Download SEAS5 dataset from ECMWF CDS API

    Args:
        output_dir: Directory to save the downloaded data
        filename_prefix: Prefix for the output filename

    Returns:
        str: Path to the downloaded file
    """
    print("Downloading SEAS5 data from ECMWF CDS...")

    # Create output directory if it doesn't exist
    os.makedirs(output_dir, exist_ok=True)

    # Current date to use in the filename
    current_date = datetime.datetime.now().strftime("%Y%m%d")
    output_file = os.path.join(output_dir,
                                f"{filename_prefix}{current_date}.grib")

    # Define the SEAS5 dataset and request parameters
    dataset = "seasonal-monthly-single-levels"
    request = {
        "originating_centre": "ecmwf",
        "system": "51",
        "variable": ["total_precipitation"],
        "year": [
            "1981", "1982", "1983", "1984", "1985", "1986", "1987", "1988",
"1989",
            "1990", "1991", "1992", "1993", "1994", "1995", "1996", "1997",
"1998",
            "1999", "2000", "2001", "2002", "2003", "2004", "2005", "2006",
"2007",
            "2008", "2009", "2010", "2011", "2012", "2013", "2014", "2015",
"2016",
            "2017", "2018", "2019", "2020", "2021", "2022", "2023", "2024", "2025"
        ],
        "month": [
            "01", "02", "03", "04", "05", "06",
            "07", "08", "09", "10", "11", "12"
        ],
        "leadtime_month": ["1", "2", "3", "4", "5", "6"],
        "data_format": "grib",
        "product_type": ["monthly_mean"],
        "area": [23, 21, -12, 53] # North, West, South, East
    }

    try:
        client = cdsapi.Client()
        client.retrieve(dataset, request, output_file)
        print(f"SEAS5 data downloaded successfully to: {output_file}")
        return output_file
    except Exception as e:
        print(f"Error downloading SEAS5 data: {e}")
        return None

```

```
def cleanup_old_seas5_files(output_dir="./data",
                             filename_prefix="seas5_precipitation_", keep_latest=True):
    """
    Remove old SEAS5 files, optionally keeping the latest one

    Args:
        output_dir: Directory containing the SEAS5 files
        filename_prefix: Prefix of the SEAS5 files
        keep_latest: Whether to keep the latest file
    """
    print("Cleaning up old SEAS5 files...")

    # Get all SEAS5 files
    pattern = os.path.join(output_dir, f"{filename_prefix}*.grib")
    files = glob.glob(pattern)

    if not files:
        print("No SEAS5 files found for cleanup.")
        return

    if keep_latest and len(files) > 1:
        # Sort files by modification time (newest last)
        files.sort(key=os.path.getmtime)
        # Remove the latest file from the list
        latest_file = files.pop()
        print(f"Keeping latest file: {os.path.basename(latest_file)}")

    # Remove remaining files
    for file in files:
        try:
            os.remove(file)
            print(f"Removed: {os.path.basename(file)}")
        except Exception as e:
            print(f"Error removing {file}: {e}")

    # Example: Download SEAS5 data and clean up old files
    seas5_file = download_seas5()
    cleanup_old_seas5_files()

    # Open the data with xarray
    import xarray as xr
    ds = xr.open_dataset(seas5_file)
    print(ds)
```

Important CDS Usage Notes:

- There are **usage limits** for data retrieval from CDS (number of requests, volume)
- Large requests are **queued** and may take time to complete
- If your request fails, the system will automatically retry
- Monitor your pending requests in the CDS web interface under “My Requests”

7.3 Data Management and Processing

The provided script includes several important features for managing the downloaded data:

1. Automatic File Management:

- CHIRPS v3.0 data is downloaded only once (checks if file already exists)

- SEAS5 files are automatically date-stamped and old files are cleaned up
- Option to keep all SEAS5 files if needed for historical analysis

2. Regional Specification:

- SEAS5 data is downloaded for a specific region ([23°N, 21°E, -12°S, 53°E])
- This focused approach reduces download size and processing time

3. Preprocessing Considerations:

After downloading both datasets, you'll typically need to:

- **Convert GRIB to NetCDF:** SEAS5 data is downloaded in GRIB format and may need conversion
- **Handle Different Spatial Resolutions:** CHIRPS and SEAS5 have different resolutions
- **Apply Temporal Alignment:** Match forecast periods with observational data

```
import xarray as xr
import cfgrib
import xesmf as xe

# Load datasets (note different formats)
forecast = xr.open_dataset("./data/seas5_precipitation_20250407.grib",
                           engine="cfgrib")
obs = xr.open_dataset("./data/chirps-v3.0.monthly.nc")

# Define the regridded if needed
regridded = xe.Regridded(
    forecast,
    obs,
    "bilinear",
    periodic=True
)

# Apply regridding
forecast_regridded = regridded(forecast)
```

This automated approach to data acquisition forms the foundation for developing robust drought monitoring applications based on the Standardized Precipitation Index (SPI).

SPI Calculation using Python Libraries

8.1 The Standardized Precipitation Index (SPI)

The Standardized Precipitation Index (SPI) serves as a primary drought indicator in the anticipatory action system, quantifying rainfall anomalies across different timescales. Originally developed by McKee et al. (1993), SPI represents how current precipitation compares to historical conditions, expressed in standard deviation units.

In the implementation found in `run-process-spi.py`, SPI is calculated for 3-month accumulation periods (SPI-3), capturing seasonal-scale rainfall anomalies particularly relevant to agricultural drought:

```
spi_3 = standardized_precipitation_index(  
    aa,  
    freq="MS",  
    window=3,  
    dist="gamma",  
    method="APP",  
    cal_start='1991-01-01',  
    cal_end='2018-01-01',  
    fitkwargs={"floc": 0}  
)
```

SPI values typically range from -4 to +4, with negative values indicating precipitation deficits. The interpretation follows standard thresholds, where values below -1 represent severe drought conditions that occur approximately once every 6-7 years, making them suitable targets for anticipatory action.

8.2 The xclim Library: Climate Indices Made Accessible

The xclim library provides a robust framework for climate data analysis, with particular strengths in calculating standardized climate indices like SPI. Developed to work seamlessly with xarray data structures, xclim implements climate indices following scientific standards while addressing computational challenges associated with large climate datasets. The details of the method is available at (https://xclim.readthedocs.io/en/v0.39.0/xclim.indicators.atmos.html#xclim.indicators.atmos._precip.standardized_precipitation_index)

For drought monitoring, xclim's `standardized_precipitation_index` function offers several critical features:

1. **Distribution Selection:** While gamma distribution is the standard choice for SPI calculation (used in our implementation), xclim supports alternative distributions including Pearson Type III and empirical distributions.

2. **Calculation Methods:** The library supports multiple calculation approaches: - "APP": Approximation method (used in our implementation) - "ML": Maximum likelihood estimation - "PWM": Probability weighted moments
3. **Calibration Period Control:** As seen in the implementation, the function allows for specific calibration periods (`cal_start` and `cal_end`), crucial for establishing a consistent reference climatology across both forecast and observation datasets.
4. **Zero Precipitation Handling:** The `fitkwargs={"floc": 0}` parameter ensures that gamma distribution fitting properly handles zero precipitation values, which can be common in arid regions.

8.3 The xarray Foundation: Multi-dimensional Data Structures

Underlying both the data processing and SPI calculation is the xarray library, which provides labeled multi-dimensional arrays designed specifically for scientific datasets. In the context of drought monitoring, xarray offers several advantages:

1. **Self-describing Data:** Coordinates, dimensions, and metadata travel with the data, simplifying the tracking of spatial and temporal information.
2. **Advanced Indexing:** The implementation leverages xarray's powerful selection capabilities (e.g., `.sel(lead=fm)`, `.sel(latitude=slice(lat_max, lat_min))`) for intuitive data subsetting.
3. **Parallel Computing Integration:** Through dask integration, xarray enables scalable processing of large ensembles, evident in the `.compute()` calls throughout the processing pipeline.
4. **Interoperability:** The seamless handling of different data formats (NetCDF, GRIB, Zarr) facilitates the integration of observation data (CHIRPS) with forecast data (SEAS5).

In the SPI calculation for ensemble forecast data, the processing handles ensemble members differently based on their characteristics. For older members (0-24), SPI is calculated using the 1991-2018 calibration period, while newer operational members (25-51) use a 2017-2024 calibration period, addressing the challenge of maintaining consistent statistical properties across the full ensemble.

The script `01-run-process-spi.py` performs four main operations:

1. Calculates SPI on CHIRPS observation data
2. Calculates SPI on SEAS51 forecast data
3. Re grids both datasets to a common grid
4. Applies a geographic mask to focus on a specific region

8.4 1. SPI Calculation on CHIRPS Data

The `process_chirps_data()` function handles the processing of CHIRPS observation data:

```
def process_chirps_data(region_id, credentials, extent, chirps_file=None,
                        output_dir='.', res=0.25):
    # Load CHIRPS data either from local file or GCS storage
    if chirps_file:
        logger.info(f"Using local CHIRPS file: {chirps_file}")
        ds = xr.open_dataset(chirps_file)
    else:
        logger.info(f"Loading CHIRPS data from GCP")
        uri = 'gs://seas51/ea_chirps_v3_monthly_20250312.zarr'
        ds = xr.open_dataset(uri, engine='zarr', consolidated=False,
```

```

storage_options={'token': credentials})

    # Subset to the region of interest
    ds_subset = ds.sel(latitude=slice(lat_min, lat_max), longitude=slice(lon_min,
lon_max))

    # Regrid to regular grid
    ds1 = ds_subset_computed.rename({'longitude': 'lon', 'latitude': 'lat'})
    dr = ds1["precip"]

    # Calculate SPI-3
    spi_3 = standardized_precipitation_index(
        aa,
        freq="MS",
        window=3,
        dist="gamma",
        method="APP",
        cal_start='1991-01-01',
        cal_end='2018-01-01',
        fitkwargs={"floc": 0}
    )

```

Key points:

- Uses xarray to handle climate data
- Applies regional subsetting to focus on the area of interest
- Calculates SPI-3 (3-month Standardized Precipitation Index) using gamma distribution
- Uses a calibration period from 1991-2018

8.5 2. SPI Calculation on SEAS51 Forecast Data

The `process_seas51_data()` function processes forecast data from ECMWF SEAS51:

```

def process_seas51_data(region_id, obs_file, credentials, extent,
    seas51_file=None, output_dir='.'):
    # Load SEAS51 data from file or GCS
    if seas51_file:
        logger.info(f"Using local SEAS51 file: {seas51_file}")
        sds = xr.open_dataset(seas51_file, engine='cfgrib',
backend_kwargs=dict(time_dims=('forecastMonth', 'time')))
    else:
        logger.info(f"Loading SEAS51 data from GCP")
        suri = 'gs://seas51/ea_seas51_20250312_v3.zarr'
        sds = xr.open_dataset(suri, engine='zarr', consolidated=False,
storage_options={'token': credentials})

    # Subset to region
    sds_subset = sds.sel(latitude=slice(lat_min, lat_max),
longitude=slice(lon_min, lon_max))

    # Calculate SPI-3 for each lead time (1-6 months)
    all_leads = []
    for lead_val in range(1, 7):
        # Process each lead time
        cont_spi = vt_apply_spi3_with_parameter_transfer(sds_subset_computed,
lead_val)
        lead_data = xr.concat(cont_spi, dim='member')
        all_leads.append(lead_data)

```

Key points:

- Processes multiple ensemble members (0-51) from the SEAS51 seasonal forecast
- Handles different lead times (1-6 months) separately
- Special handling for members 0-24 vs 25-51 with different calibration periods
- Concatenates multiple ensemble members into a single dataset

8.6 3. Regridding Operations

Both datasets need to be on the same grid for comparison. The script includes regridding operations:

```
# For CHIRPS data:
ds_out = xr.Dataset({
    "lat": ([ "lat" ], np.arange(lat_min, lat_max, res), {"units":
    "degrees_north"}),
    "lon": ([ "lon" ], np.arange(lon_min, lon_max, res), {"units": "degrees_east"}),
})
regridder = xe.Regridder(ds1, ds_out, "conservative")
dr_out = regridder(dr, keep_attrs=True)

# For SEAS51 data (to match CHIRPS grid):
ds_out = xr.Dataset({
    "lat": ([ "lat" ], kn_obs['lat'].values, {"units": "degrees_north"}),
    "lon": ([ "lon" ], kn_obs['lon'].values, {"units": "degrees_east"}),
})
regridder = xe.Regridder(gd2, ds_out, "bilinear", periodic=False)
dr_out = regridder(agd, keep_attrs=True)
```

Key points:

- Uses xESMF for regridding operations
- CHIRPS data uses conservative regridding to a regular grid
- SEAS51 forecast data is regridded to match the CHIRPS observations grid using bilinear interpolation
- Regular grid uses the specified resolution (default 0.25°)

8.7 4. Masking Operations

The `mask_netcdf_with_shapefile()` function applies a geographic mask to both datasets:

```
def mask_netcdf_with_shapefile(forecast_path, obs_path, shapefile_df,
                               buffer_size=0.25,
                               output_forecast_path='masked_forecast.nc',
                               output_obs_path='masked_obs.nc'):

    # Load datasets
    forecast_ds = xr.open_dataset(forecast_path)
    obs_ds = xr.open_dataset(obs_path)

    # Create buffered geometry if needed
    buffered_gdf = gdf.copy()
    if buffer_size > 0:
        buffered_gdf['geometry'] = gdf.geometry.buffer(buffer_size)

    # Create mask from shapefile
    f_mask = regionmask.mask_geopandas(buffered_gdf.geometry, f_lons, f_lats)
    f_bool_mask = ~np.isnan(f_mask)
```

```

# Expand mask to match dataset dimensions
f_expanded_mask = f_bool_mask.expand_dims({
    "lead": forecast_ds.lead,
    "member": forecast_ds.member,
    "init": forecast_ds.init
})

# Apply mask to set values outside region to NaN
masked_forecast = forecast_ds.copy(deep=True)
for var in forecast_ds.data_vars:
    masked_forecast[var] = forecast_ds[var].where(f_expanded_mask, np.nan)

```

Key points:

- Uses regionmask to create masks from GeoDataFrame
- Applies a buffer around the region to include border areas
- Expands the 2D mask to match the dimensionality of each dataset
- Sets values outside the mask region to NaN
- Creates masked versions of both the observations and forecasts

8.8 Main Execution Flow

The script's main execution follows this sequence:

```

if __name__ == "__main__":
    # Define region and get extent
    region_id = 'kmj'
    gdf, extent = get_region_bounds(region_id, use_local=True)

    # Process CHIRPS data
    obs_file = process_chirps_data(
        region_id, credentials, extent, chirps_file=chirps_file
    )

    # Process SEAS51 data
    fct_file = process_seas51_data(
        region_id, obs_file, credentials, extent, seas51_file=seas51_file
    )

    # Apply masking
    masked_fct, masked_obs = mask_netcdf_with_shapefile(
        forecast_path=fct_file,
        obs_path=obs_file,
        shapefile_df=shapefile_df,
        buffer_size=0.25
    )

```

This comprehensive processing pipeline takes raw climate data and prepares it for drought forecasting analysis by:

1. Loading and subsetting the data to the region of interest
2. Computing SPI values for both observations and forecasts
3. Ensuring both datasets are on the same spatial grid
4. Applying geographic masks to focus only on the relevant area

The resulting NetCDF files can then be used for further analysis and visualization of drought risks and forecasting skill assessment.

Regridding Datasets with xESMF

9.1 Spatial Resolution Harmonization

A critical step in comparing forecasts with observations involves addressing the inherent resolution mismatch between datasets. In the drought anticipatory action system, CHIRPS observations (native resolution of 0.05°) and SEAS5 forecasts (native resolution of 1.0°) must be harmonized to a common grid to enable meaningful comparisons. The `run-process-spi.py` script implements this harmonization through regridding, bringing both datasets to an intermediate resolution of 0.25° (approximately 25 km at the equator).

The regridding process handles two distinct cases:

1. CHIRPS Observation Regridding (5 km $\rightarrow$ 25 km):

```
# Create output grid with regular spacing
ds_out = xr.Dataset({
    "lat": ([ "lat" ], np.arange(lat_min, lat_max, res), {"units":
    "degrees_north"}),
    "lon": ([ "lon" ], np.arange(lon_min, lon_max, res), {"units":
    "degrees_east"}),
})

# Regrid using conservative method
regridder = xe.Regridder(ds1, ds_out, "conservative")
dr_out = regridder(dr, keep_attrs=True)
```

This upscaling process preserves precipitation totals through conservative regridding, which maintains the area-integrated rainfall quantity during the transition to a coarser grid.

2. SEAS5 Forecast Regridding (100 km $\rightarrow$ 25 km):

```
# Create output grid matching observations
ds_out = xr.Dataset({
    "lat": ([ "lat" ], kn_obs['lat'].values, {"units": "degrees_north"}),
    "lon": ([ "lon" ], kn_obs['lon'].values, {"units": "degrees_east"}),
})

# Perform regridding
regridder = xe.Regridder(gd2, ds_out, "bilinear", periodic=False)
dr_out = regridder(agd, keep_attrs=True)
```

For the forecast data, bilinear interpolation is employed to downscale from 100 km to 25 km, producing smoother transitions between grid cells appropriate for the forecast uncertainty.

9.2 The xESMF Library

The Earth System Modeling Framework (ESMF) regridding functionality is made accessible through xESMF, a Python interface that integrates seamlessly with xarray datasets. This library provides multiple interpolation methods appropriate for different variables and scientific needs:

1. **Conservative:** Preserves the integral of the field (used for CHIRPS precipitation upscaling)
2. **Bilinear:** Weighted average of nearest grid points (used for SEAS5 interpolation)
3. **Nearest-neighbor:** Assigns value from nearest source point
4. **Higher-order patch:** Smoother interpolation useful for continuous fields

In the implementation, regridding objects are created once and stored for repeated application across dimensions:

```
regridder = xe.Regriddler(gd2, ds_out, "bilinear", periodic=False)
```

The `periodic=False` parameter indicates that the longitude dimension is not treated as periodic (wrapped around the globe), which is appropriate for regional domains in East Africa.

9.3 Handling Special Cases in Regridding

The regridding implementation addresses several technical challenges:

1. **Coordinate System Compatibility:** The script ensures coordinate naming consistency before regridding:

```
# Rename coordinates for consistency if needed
if 'longitude' in ds_p_m1.dims and 'latitude' in ds_p_m1.dims:
    gd2 = ds_p_m1.rename({'longitude': 'lon', 'latitude': 'lat'})
```

2. **Land-Sea Mask Preservation:** The regridding process preserves missing values to maintain the integrity of land-sea boundaries.
3. **Ensemble Dimension Handling:** When processing SEAS5 ensemble data, regridding is applied individually to each lead time to manage memory efficiently:

```
for fm in range(min(6, len(all_leads))):
    # Regridding for each lead time
    # ...
```

This approach ensures that both datasets align on a common 0.25° grid, enabling direct comparison between forecasts and observations while balancing spatial detail against computational efficiency in the SPI calculation and verification process.

Aligning Forecast and Observation Data with climpred

10.1 Time Alignment Challenge in Forecast Verification

One of the most complex challenges in verifying seasonal forecasts is aligning the time coordinates between forecasts and observations. The `make_obs_fct_dataset` function in `vthree_utils.py` addresses this critical task by integrating the SEAS5 forecast and CHIRPS observation datasets with different temporal structures:

```
def make_obs_fct_dataset (params) :
    """
    Prepares observed and forecasted dataset subsets for a specific region,
    season, and lead time by aligning their temporal coordinates.
    """
```

The function performs a sophisticated alignment process that accounts for three distinct time dimensions in seasonal forecasting:

1. **Initialization Time (init):** When the forecast was issued
2. **Lead Time (lead):** How many months into the future the forecast extends
3. **Valid Time:** When the forecast is valid for (init + lead)

Since CHIRPS observations have only a `time` dimension while SEAS5 forecasts have both `init` and `forecastMonth` (lead time), direct comparison requires creating a compatible coordinate system.

10.2 The climpred Library

The alignment utilizes `climpred`, a specialized Python library for seasonal forecast verification that bridges the gap between forecast and observational data structures:

```
hindcast = HindcastEnsemble(a_fc)
hindcast = hindcast.add_observations(a_obs)
a_fcl = hindcast.get_initialized()
```

This sequence of operations performs several crucial tasks:

1. Creates a `HindcastEnsemble` object from the forecast dataset
2. Adds the observation dataset as a reference
3. Computes valid times for each forecast initialization and lead time combination
4. Realigns the forecast dataset with explicit `valid_time` coordinates

The `HindcastEnsemble` framework enables consistent handling of forecast lead times across multiple initialization dates, crucial for calculating skill metrics.

10.3 Lead Time Calculation Example: JJA Season SPI

To illustrate the lead time concept, consider forecasting the June-July-August (JJA) SPI:

Forecast Month	Lead Time	Valid Period (SPI JJA)	Practical Usage
March	Lead time 4	June-July-August	Early planning
April	Lead time 3	June-July-August	Refinement
May	Lead time 2	June-July-August	Final decision

For example, forecasts initialized in March (Lead time 4) need three months to reach the start of the JJA period, providing earlier warning but typically with lower skill. Conversely, May forecasts (Lead time 2) offer higher accuracy but less preparation time.

The alignment process must account for these varying lead times, extracting the correct forecast-observation pairs:

```
# Select forecast for specific lead time
a_fc2 = a_fc1.isel(lead=params.lead_int)

# Find common dates between forecasts and observations
common_dates = np.intersect1d(fct_valid_times, obs_times)

# Filter observations to include only dates with matching forecasts
a_obs3 = a_obs2.sel(time=common_dates)

# Calculate initial dates by subtracting lead time from valid dates
a_fc3_init_dates = common_dates.astype("datetime64[M]") - np.timedelta64(
    int(a_fc3.lead.values), "M"
)

# Select forecasts initialized on these dates
a_fc4 = a_fc3.sel(init=a_fc3_init_dates)
```

This process creates perfect temporal alignment between forecast valid times and observation times, enabling direct comparison for verification statistics. It ensures that when calculating metrics like hit rates or false alarm ratios, the system is comparing forecasts and observations for exactly the same temporal periods.

10.4 Seasonal SPI Product Identification

Another sophisticated aspect of the alignment is identifying specific SPI seasonal products from the time coordinates:

```
# Create season identifiers (e.g., "MAM", "JJA")
spi_prod_list = spi3_prod_name_creator(a_fc2, "valid_time")
obs_spi_prod_list = spi3_prod_name_creator(a_obs, "time")

# Assign season identifiers as coordinates
a_fc2 = a_fc2.assign_coords(spi_prod=("init", spi_prod_list))
a_obs1 = a_obs.assign_coords(spi_prod=("time", obs_spi_prod_list))

# Filter to the specified season
a_fc3 = a_fc2.where(a_fc2.spi_prod == params.season_str, drop=True)
a_obs2 = a_obs1.where(a_obs1.spi_prod == params.season_str, drop=True)
```

This approach enables targeting specific three-month periods like MAM (March-April-May) or JJA (June-July-August), aligning with key agricultural seasons in East Africa while maintaining the temporal correspondence between forecast and observation datasets.

Forecast Verification with xhistogram

11.1 Multi-dimensional Contingency Analysis

Forecast verification forms the cornerstone of any anticipatory action system, providing evidence that the forecasting system can reliably detect drought events before they occur. The `vthree_utils.py` script implements comprehensive verification methodology using the `xhistogram` library (imported as `xhist`), which enables efficient calculation of contingency tables and skill metrics across multiple dimensions.

At the core of this verification approach is the `xhist_metrics_2d` function, which performs grid-based verification of forecast probabilities against binary drought observations:

```
def xhist_metrics_2d(obs_data, ens_prob_data, params, calculate_auroc=True):
    """
    Calculates verification metrics for ensemble drought forecasts using
    multi-dimensional contingency tables.
    """
    # Get threshold for the specified region and season
    threshold_dict = get_threshold(params.region_id, params.sc_season_str)
    threshold = threshold_dict[params.level]

    # Process trigger values and calculate metrics
    df = process_trigger_values(
        obs_data, ens_prob_data, params, threshold, calculate_auroc
    )

    return df
```

This approach leverages multi-dimensional histograms to efficiently calculate contingency tables across the entire spatial domain, without requiring explicit loops over each grid cell.

11.2 Constructing Contingency Tables

The key innovation in this verification framework is the use of `xarray`'s histogram functionality to construct contingency tables directly. This is implemented in the `calculate_contingency_table` function:

```
def calculate_contingency_table(obs_event, forecast_event, params):
    """Creates a 2x2 contingency table using xhistogram"""
    obs_event1 = obs_event[params.spi_prod_name]
    forecast_event1 = forecast_event[params.spi_prod_name]

    # Align forecast to observation time
```

```

forecast_event3 = forecast_event1.assign_coords(
    init=forecast_event1.coords["valid_time"]
)
forecast_event3 = forecast_event3.rename({"init": "time"})

# Histogram calculation produces the contingency table
contingency_table = xhist.histogram(
    obs_event1, forecast_event3, bins=[2, 2], density=False, dim=["lat",
"lon"]
)

return contingency_table

```

This function:

1. Extracts the relevant SPI values from the observation and forecast datasets
2. Aligns the temporal coordinates of forecasts with observations
3. Computes a 2D histogram where each axis represents binary values (0=no drought, 1=drought)
4. Returns a contingency table with dimensions [time, obs_event, forecast_event]

The resulting contingency table has four elements representing the classic “hits”, “misses”, “false alarms”, and “correct negatives” for each time step.

11.3 Contingency Tables and Forecast Verification in Drought Anticipatory Action

11.4 Integrating Observation and Forecast Data

In drought early warning systems, the process of combining observations and forecasts involves several key steps:

1. **Data Preparation:** Both observed (CHIRPS) and forecasted (SEAS51) data are processed to calculate SPI (Standardized Precipitation Index) values.
2. **Threshold Application:** A threshold (typically $SPI \leq -1$ for severe drought) is applied to both datasets to create binary events.
3. **Probability Extraction:** For forecasts, the percentage of ensemble members exceeding the threshold is calculated as the forecast probability.
4. **Trigger Value Testing:** Different probability thresholds are tested to determine optimal trigger values for initiating anticipatory actions.

11.5 Understanding Contingency Tables

A contingency table is a 2×2 matrix that summarizes the relationship between observed events and forecasted events:

Table 11.1. Contingency Table

	Observed: Yes	Observed: No
Forecasted: Yes	a (Hit)	b (False Alarm)
Forecasted: No	c (Miss)	d (Correct Rejection)

Where:

- **a (Hit):** Drought was forecasted and occurred
- **b (False Alarm):** Drought was forecasted but did not occur
- **c (Miss):** Drought was not forecasted but occurred
- **d (Correct Rejection):** Drought was not forecasted and did not occur

In the code, this table is generated by comparing binary observations (observed SPI $\leq$ threshold) with binary forecasts (forecast probability $\geq$ trigger value).

11.6 Verification Metrics Explained

From the contingency tables, a comprehensive set of verification metrics is calculated to assess different aspects of forecast quality:

1. Hit Rate (HR):

$$HR = \frac{\text{hits}}{\text{hits} + \text{misses}}$$

Measures the proportion of observed drought events that were correctly forecast. Also called Probability of Detection (POD). Ranges from 0 (worst) to 1 (perfect).

2. False Alarm Ratio (FAR):

$$FAR = \frac{\text{false_alarms}}{\text{false_alarms} + \text{hits}}$$

Measures the proportion of drought warnings that were false alarms. Ranges from 0 (perfect) to 1 (worst).

3. Bias Score (BS):

$$BS = \frac{\text{hits} + \text{false_alarms}}{\text{hits} + \text{misses}}$$

Indicates whether the system tends to under-forecast ($BS < 1$) or over-forecast ($BS > 1$) drought events.

4. Hanssen-Kuipers Score (HK):

$$HK = HR - \frac{\text{false_alarms}}{\text{false_alarms} + \text{correct_negatives}}$$

Also known as the True Skill Statistic (TSS), measures the ability to discriminate between drought and non-drought events. Ranges from -1 to 1.

5. Heidke Skill Score (HSS):

$$HSS = \frac{2(\text{hits} \cdot \text{correct_negatives} - \text{misses} \cdot \text{false_alarms})}{(\text{hits} + \text{misses})(\text{misses} + \text{correct_negatives}) + (\text{hits} + \text{false_alarms})(\text{false_alarms} + \text{correct_negatives})}$$

Measures the improvement of the forecast over random chance. Ranges from $-\infty$ to 1.

6. Area Under ROC Curve (AUROC):

Measures the forecast's ability to discriminate between drought and non-drought events across all possible threshold values. Ranges from 0 to 1, with 0.5 indicating no skill.

11.7 AUROC and Bootstrapping

The AUROC calculation with bootstrapping represents the most sophisticated element of the verification methodology:

```

def calculate_auroc_score(contingency_table, trigger_value, n_bootstrap=1000):
    """
    Calculates the Area Under the ROC curve with confidence intervals
    using bootstrapping.
    """
    # Sum the contingency table across time
    summed_table = contingency_table.sum(dim="time")

    # Extract counts
    ct = summed_table.values.flatten()
    correct_negatives, false_alarms, misses, hits = ct
    total = hits + false_alarms + misses + correct_negatives

    # Perform bootstrapping
    auroc_bootstrap_scores = []
    for _ in range(n_bootstrap):
        bootstrap_counts = np.random.multinomial(
            total,
            [
                hits / total,
                misses / total,
                false_alarms / total,
                correct_negatives / total,
            ],
            size=1,
        )

        bootstrap_hits, bootstrap_misses, bootstrap_false_alarms,
        bootstrap_correct_negatives = bootstrap_counts[0]

        auroc_bootstrap_scores.append(
            calculate_auroc(
                bootstrap_hits,
                bootstrap_misses,
                bootstrap_false_alarms,
                bootstrap_correct_negatives,
            )
        )

    # Calculate mean and confidence intervals
    auroc_score = np.mean(auroc_bootstrap_scores)
    auroc_lb, auroc_ub = np.percentile(auroc_bootstrap_scores, [2.5, 97.5])

    return {
        "trigger_value": trigger_value,
        "auroc_score": auroc_score,
        "auroc_lb": auroc_lb,
        "auroc_ub": auroc_ub,
    }

```

This bootstrap approach:

1. Resamples from the original contingency table 1000 times
2. Calculates the AUROC score for each resample
3. Computes the mean AUROC score and 95% confidence intervals

Bootstrapping is crucial for drought verification because:

- Droughts are relatively rare events, leading to sampling uncertainty
- Confidence intervals provide stakeholders with a measure of uncertainty
- It helps identify whether skill scores are statistically significant

11.8 2D vs 1D Verification Methods

The system implements two complementary verification approaches:

1. 2D Verification (Grid-based):

```
def run_xhist2d(params):
    """Grid-based verification across the entire spatial domain"""
    # ...
    df = xhist_metrics_2d(obs_data, fct_mod, params, calculate_auroc=True)
    # ...
```

This approach calculates metrics at each grid point and then aggregates the results, preserving the spatial structure of forecast skill. It's useful for understanding regional variations in forecast quality.

2. 1D Verification (Region-based):

```
def run_xhist1d(params):
    """Region-based verification using spatial means"""
    # ...
    obs_df = mean_obs_spi(obs_data, params.spi_prod_name)
    fct_df = mean_emp_prob(fct_mod, fct_sev, fct_ext, params.spi_prod_name)
    # ...
```

This approach first calculates spatial means across the region and then performs verification on these mean values. It's useful for making operational decisions at the regional level.

Both approaches are necessary because:

- 2D verification captures the spatial heterogeneity of forecast skill
- 1D verification aligns better with decision-making at administrative levels
- Comparing both can identify issues with compensating errors in the forecasting system

The comprehensive verification framework implemented in these scripts provides the statistical basis for selecting optimal trigger values for anticipatory action, balancing the desire for high hit rates against the need to minimize false alarms within operational constraints.

11.9 CSV Output Structure

The CSV output file contains verification metrics calculated from the contingency table. Here's an explanation of each column:

Table 11.2. CSV Output Columns

Column Name	Description
Unnamed: 0, index	Row identifiers
x2d_region	Region identifier (e.g., 0 for Karamoja)
x2d_leadtime	Lead time in months (forecast horizon)
x2d_season	Season being forecasted (e.g., "mam" for March-April-May)
x2d_level	Drought severity level (mild, mod, sev/ext)
trigger_value	Probability threshold value being tested (0-100%)
time_step	Specific time step or window being analyzed
hit_rate	Proportion of observed events that were correctly forecasted ($a/(a+c)$)
false_alarm_ratio	Proportion of false alarms in all forecasts ($b/(a+b)$)

bias_score	Ratio of forecasted to observed events $((a+b)/(a+c))$
hanssen_kuipers_score	Difference between hit rate and false alarm rate $(a/(a+c) - b/(b+d))$
heidke_skill_score	Accuracy relative to random chance
total	Total number of cases $(a+b+c+d)$
auroc_score	Area Under ROC curve - measures discrimination ability
auroc_lb, auroc_ub	Lower and upper confidence bounds for AUROC
cat	Drought category (mild, mod, sev/ext)
obs_count	Number of observed drought events
threshold	SPI threshold value used for event definition
hits	Number of hits (a)
misses	Number of misses (c)
FA	Number of false alarms (b)
CN	Number of correct negatives (d)
hit_percentage	Percentage of observed events that were correctly forecasted

The run script verification

The script `02-run-fv-kmj.py` implements a comprehensive drought forecast verification system following the methodology outlined in the works of Guimarães Nobre et al. The script systematically processes seasonal forecast data to identify reliable triggers for anticipatory action.

12.1 Execution Flow

The script follows this sequence:

```
import os
from dotenv import load_dotenv
import numpy as np
import pandas as pd
import altair as alt
from io import StringIO
import xarray as xr
from sklearn.metrics import roc_auc_score
from xbootstrap import block_bootstrap
import xskillscore as xs

from dask.distributed import Client
from vthree_utils import get_threshold
from vthree_utils import BinCreateParams
from vthree_utils import run_xhist2d
from vthree_utils import run_xhist1d

# Load environment variables and create parameter object
params = BinCreateParams(
    region_id=0,
    season_str="MAM", # March-April-May season
    lead_int=2, # Lead time in months
    level="mod", # Moderate drought category
    region_name_dict={0: "Karamoja", 1: "Marsabit", 2: "Wajir"},
    spi_prod_name="spi3", # 3-month SPI
    data_path=ea_input_path,
    output_path=os.path.join(os.getcwd(), "output"),
    spi4_data_path="",
    obs_netcdf_file=os.path.join(ea_input_path, "kn_obs_spi3_20240717.nc"),
    fct_netcdf_file=os.path.join(ea_input_path, "kn_fct_spi3_20240717.nc"),
    service_account_json=sa_file,
    gcs_file_url=polygon_pq_uri,
    region_filter="kmj",
)

# First perform 2D verification for different lead times and drought levels
run_xhist2d(params)
```

```
run_xhist1d(params)

params.lead_int = 3
run_xhist2d(params)
run_xhist1d(params)

params.lead_int = 4
run_xhist2d(params)
run_xhist1d(params)

# Change season to JJA (June-July-August) and repeat
params.season_str = "JJA"
params.sc_season_str = "jja"
params.lead_int = 2
run_xhist2d(params)
run_xhist1d(params)
# And so on...
```

12.2 2D Verification with run_xhist2d()

The 2D verification examines the relationship between ensemble forecast probabilities and observed drought events across the spatial domain for each threshold category.

```
def run_xhist2d(params):
    threshold_dict = get_threshold(params.region_id, params.sc_season_str)
    obs_data, ens_data = make_obs_fct_dataset(params)
    fct_mod, fct_sev, fct_ext = seas5l_patch_empirical_probability(
        ens_data, threshold_dict
    )
```

- One dimension is observed drought events (True/False)
- Second dimension is forecasted drought events (True/False)

For each possible trigger value (probability threshold), it calculates:

- Hit rate
- False alarm ratio
- Bias score
- Hanssen-Kuipers score
- Heidke skill score
- AUROC score (with bootstrapping for uncertainty estimation)

12.3 1D Verification with run_xhist1d()

After 2D analysis, the script performs 1D verification using the pre-filtered values from 2D verification:

```
def run_xhist1d(params):
    threshold_dict = get_threshold(params.region_id, params.sc_season_str)
    obs_data, ens_data = make_obs_fct_dataset(params)
    fct_mod, fct_sev, fct_ext = seas5l_patch_empirical_probability(
        ens_data, threshold_dict
    )
```

This 1D verification focuses on area-averaged time series for both observations and forecasts, evaluating different trigger values for their performance in capturing drought events.

12.4 Key Functions from `vthree_utils.py`

The script relies on several critical functions from `vthree_utils.py`:

12.4.1 `get_threshold()`

Retrieves predefined drought thresholds for each region and season

12.4.2 `make_obs_fct_dataset()`

Aligns observational and forecast datasets with correct:

- Region selection
- Season filtering
- Lead time selection
- Time coordinate alignment

12.4.3 `seas51_patch_empirical_probability()`

Calculates empirical probabilities of exceeding thresholds from ensemble forecasts, handling the transition from 25 members (1981-2017) to 51 members (2017-present)

12.4.4 `xhist_metrics_2d()`

Creates 2D contingency tables and evaluates metrics for each possible trigger value

12.4.5 `chosen_triggers_metrix()`

Calculates 1D verification metrics for specific trigger values:

- Hit rate and percentage
- False alarm ratio
- Bias score
- AUROC score

12.4.6 `update_ctdb()`

Updates the filtered contingency table database from 2D analysis with 1D analysis metrics

12.4.7 `trigger_decision_dict()`

Selects optimal triggers based on hit rate, false alarm ratio, and other metrics

12.5 Detailed Verification Process

The verification process follows these steps:

1. Data Preparation:

- Load observation data (CHIRPS)
- Load forecast data (SEAS5.1) for different lead times

- Calculate SPI (Standardized Precipitation Index)
- Apply thresholds to define drought events
- Calculate ensemble probabilities

2. 2D Spatial Verification (`run_xhist2d`):

- Create 2D contingency tables using `xhistogram.xarray`
- Calculate verification metrics for each probability trigger
- Filter for triggers meeting quality criteria (AUROC > 0.5, hit rate > 0.65, FAR < 0.35)
- Save filtered results to CSV files

3. 1D Time Series Verification (`run_xhist1d`):

- Calculate area-mean SPI values
- Apply 1D verification to triggers identified in 2D analysis
- Update metrics with 1D performance statistics
- Generate final verification metrics

4. Decision Support:

- Identify optimal triggers based on verification metrics
- Create visualizations and tables for decision-makers
- Document performance statistics for each selected trigger

This systematic approach ensures comprehensive evaluation of forecast skill and reliable trigger identification for anticipatory action.

Trigger Selection for Anticipatory Action against Droughts in East Africa

13.1 Forecast Verification and Trigger Selection Process

13.1.1 Data Processing Pipeline

1. Initial Processing

- CHIRPS rainfall observations are processed to calculate SPI-3
- SEAS5 precipitation forecasts are processed to calculate SPI-3 for different lead times
- Both datasets are spatially aligned and masked to the region of interest (e.g., Karamoja)

2. Verification Workflow

- `run_xhist2d.py` performs 2D verification by comparing ensemble SPI forecasts against observed SPI
- `run_xhist1d.py` performs 1D verification by comparing mean ensemble SPI against regional mean SPI
- Results are saved as CSV files for different lead times, regions, and seasons

13.1.2 Critical Manual Step: Trigger Selection

The most critical manual step in the workflow is **trigger selection**, which occurs after verification metrics are calculated. This process:

1. Examines the CSV files generated by `run_xhist2d.py` and `run_xhist1d.py`
2. Identifies optimal trigger values based on verification metrics
3. Manually marks selected triggers by adding a value of '1' in the `td` column

For example, in the CSV file for Karamoja MAM season with lead time 3:

```
tv year# hr far bs hk hs au odc h m FA CN h% cat td
15.2 10 96.9 1.6 1.0 -0.0 -0.0 1.0 13 12 1 27 2 92.3 mild 0.0
...
68.0 21 72.7 0.0 0.7 nan 0.0 0.7 8 3 5 17 17 37.5 mod 1.0
...
```

The row with `td=1.0` indicates the manually selected trigger value (68% probability) for the moderate category.

13.1.3 Selection Criteria

Triggers are selected based on a combination of metrics:

- **AUROC** (Area Under ROC Curve) > 0.5 (skill better than random)
- **Hit Rate** (HR) > 50% (minimum detection rate)
- **False Alarm Ratio** (FAR) < 35% (maximum acceptable false alarms)
- **Operational considerations** (lead time, practical implementation)
- **Trigger Decision:** From candidates meeting the above criteria, select the trigger value that:
 - Maximizes hit rate (if focusing on preventing missed events)
 - Minimizes false alarm ratio (if false alarms are costly)
 - Achieves bias score closest to 1 (for unbiased forecasts)

The CSV output contains all candidate trigger values, with the preferred trigger often highlighted or marked (typically with a `td` column value of 1).

13.2 Verification Metrics

13.2.1 2D Verification Metrics

- Used for spatial analysis of forecast skill
- Calculates contingency table statistics by gridcell
- Produces `auroc_scores`, `hit_rates`, and `false_alarm_ratios`

13.2.2 1D Verification Metrics

- Analyzes regional mean forecast performance
- Used for operational decision-making
- Focuses on practical application of triggers
- Metrics include:
 - `hit_rate`: Percent of actual drought events correctly forecast
 - `false_alarm_ratio`: Proportion of forecast events that didn't occur
 - `bias_score`: Ratio of forecast frequency to observed frequency
 - `hanssen_kuipers_scores`: Difference between hit rate and false alarm rate
 - `heidke_skill_scores`: Forecast accuracy relative to random chance

13.3 Implementation in the Ready, Set & Go! System

The Ready, Set & Go! system uses this verification process to:

1. **Ready Phase:** Use longer lead time forecasts with acceptable skill for early preparedness.
2. **Set Phase:** Use shorter lead time forecasts with higher skill for confirmation.
3. **Go! Phase:** Implement anticipatory actions based on both forecasts exceeding their respective triggers.

This double-confirmation approach balances the need for advance notice with forecast reliability, typically achieving higher hit rates and lower false alarm ratios than a single trigger approach.

13.4 Example of Analysis Using the CSV Data

Here's how the CSV data might be used in decision-making:

```
# Find the trigger with highest hit rate among candidates with FAR < 35%
candidates = df[(df['auroc_score'] > 0.5) & (df['hit_rate'] > 50) &
(df['false_alarm_ratio'] < 35)]
best_trigger = candidates.loc[candidates['hit_rate'].idxmax()]

# Or find the trigger with bias score closest to 1 (most unbiased)
best_unbiased = candidates.iloc[(candidates['bias_score'] - 1).abs().argsort()[0]]

# Relevant metrics for decision-makers
print(f"Recommended trigger value: {best_trigger['trigger_value']}%")
print(f"Hit rate: {best_trigger['hit_rate']:.1f}%")
print(f"False alarm ratio: {best_trigger['false_alarm_ratio']:.1f}%")
print(f"Expected return period: Once every {total/(hits+FA):.1f} years")
```

This analytical process helps stakeholders select appropriate trigger values for implementing anticipatory actions, balancing the costs of action against the benefits of early preparation for drought events.

13.5 Downstream Processing Dependencies

The manually selected triggers (marked with `td=1`) are essential inputs for:

1. `run_bar_plot.py`
 - Creates comparative bar visualizations
 - Shows observed SPI values alongside forecast probabilities
 - Highlights performance based on selected trigger values
2. `run_heatmap.py`
 - Generates heatmaps of hit rates and false alarm ratios
 - Visualizes performance across different months and lead times
 - Supports seasonal decision-making
3. `run_latex_table.py`
 - Produces formatted tables of available triggers
 - Highlights selected triggers
 - Documents full decision metrics for stakeholders

13.6 Forecast Skill Assessment

For each potential trigger, the system calculates:

- Number of hits (h): Correctly forecast drought events
- Number of misses (m): Unforecast drought events
- Number of false alarms (FA): Forecast events that didn't occur

- Number of correct negatives (CN): Correctly forecast non-events
- Hit percentage (h%): Percentage of observed events successfully forecast

13.7 Operational Implementation

The selected triggers enable a three-phase anticipatory action approach:

1. **Ready:** Initial alert based on early forecasts
2. **Set:** Confirmation with subsequent forecasts
3. **Go:** Implementation of anticipatory actions

The triggers serve as decision thresholds for initiating the anticipatory action workflow, with different trigger values potentially applied based on vulnerability levels (general vs. emergency menus).

13.8 Time Window Considerations

Triggers are optimized for specific seasonal windows:

- **Window 1:** Start to mid-season (e.g., MAM for Karamoja)
- **Window 2:** Middle to end of season (e.g., JJA for Karamoja)

This temporal approach allows for targeted interventions at critical points in the agricultural cycle.

Map Visualization for Ensemble Forecasts and Observations

This document explains how to create map visualizations from ensemble forecasts and observational data for drought monitoring using Python GIS tools.

14.1 Setting Up the Environment

First, import the necessary libraries:

```
import xarray as xr
import numpy as np
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import geopandas as gpd
import os
from PIL import Image
from shapely import wkb
```

14.2 Data Organization with DataTree

DataTree provides a hierarchical structure for organizing ensemble data efficiently:

```
def helper_stamp_plot(ens_data, obs_data, fct_mod, fct_sev, fct_ext):
    """
    Helper function to organize data into a tree structure.

    Parameters:
    ens_data (xarray.Dataset): Ensemble data
    obs_data (xarray.Dataset): Observation data
    fct_mod, fct_sev, fct_ext (xarray.Dataset): Forecast datasets
    """
    seas51tree = xr.DataTree()

    # Add ensemble members
    for member in ens_data.member:
        member_data = ens_data.sel(member=member)
        seas51tree[f"ensemble/member_{int(member)}"] = xr.DataTree(
            name=f"member_{int(member)}",
            dataset=member_data
        )
```

Add observation and forecast datasets to the DataTree:

```
# Add observation and forecast datasets
```

```

seas5ltree["observation"] = xr.DataTree(
    name="observation",
    dataset=obs_data
)
seas5ltree["fct_mod"] = xr.DataTree(
    name="fct_mod",
    dataset=fct_mod
)
seas5ltree["fct_sev"] = xr.DataTree(
    name="fct_sev",
    dataset=fct_sev
)
seas5ltree["fct_ext"] = xr.DataTree(
    name="fct_ext",
    dataset=fct_ext
)

return seas5ltree

```

14.3 Creating Single-Row Time Series Plot

The following function creates visualizations for a specific initialization date:

```

def create_single_row_plot(tree, init, params, region_geom):
    """
    Create a row of maps for a specific initialization date.

    Parameters:
    tree (xarray.DataTree): Data organized in tree structure
    init (datetime): Initialization date
    params (object): Parameters for plotting
    region_geom (geometry): Region geometry for overlay
    """

    members = list(tree["ensemble"].children.keys())
    valid_times = tree["ensemble/member_0"].ds.valid_time.values
    num_members = len(members)
    num_additional_plots = 4 # Obs, mod, sev, ext
    total_plots = num_members + num_additional_plots

```

Set up the figure and define color scales:

```

# Create figure with map projections
fig, axs = plt.subplots(
    1, total_plots,
    figsize=(2 * total_plots, 2),
    subplot_kw={"projection": ccrs.PlateCarree()})

# Define color scales
ensemble_cmap_range = (-4, 4)
fct_cmap_range = (0.0, 1.0)

# Get valid time for this initialization
valid_time = valid_times[
    np.where(tree["ensemble/member_0"].ds.init.values == init)[0][0]
]

```

Plot ensemble members and additional data:

```

# Plot ensemble members

```

```

for j, member_key in enumerate(members):
    _plot_ensemble_member(
        tree, member_key, params.spi_prod_name, init,
        axs[j], ensemble_cmap_range, region_geom,
        shape_overlay=(j == 0)  # Only show shapefile on first plot
    )

# Add observation and forecast plots
plot_titles = ["Obs", "mod", "sev", "ext"]
plot_keys = ["observation", "fct_mod", "fct_sev", "fct_ext"]

for k, (title, key) in enumerate(zip(plot_titles, plot_keys)):
    _plot_additional_data(
        tree, key, params.spi_prod_name, init, valid_time,
        axs[num_members + k], title,
        ensemble_cmap_range if key == "observation" else fct_cmap_range
    )

return fig, axs, (2 * total_plots, 2)

```

14.4 Plotting Map Elements

The system uses helper functions to plot ensemble members:

```

def _plot_ensemble_member(tree, member_key, variable, init, ax, cmap_range,
                          geom, shape_overlay=False):
    """Plot individual ensemble member."""
    member_data = tree[f"ensemble/{member_key}"].ds[variable]
    data = member_data.sel(init=init).values
    time_np64 = np.array(str(init), dtype="datetime64[ns]")
    year = pd.to_datetime(time_np64).year

    # Plot data
    ax.pcolormesh(
        tree["ensemble/member_0"].ds.lon.values,
        tree["ensemble/member_0"].ds.lat.values,
        data, cmap="RdBu", transform=ccrs.PlateCarree(),
        vmin=cmap_range[0], vmax=cmap_range[1]
    )

```

Add title, formatting, and optional shapefile overlay:

```

# Add title and formatting
ax.set_title(f'{year}m{member_key.split("_")[1]}', fontsize=6)
ax.set_xticks([])
ax.set_yticks([])

# Overlay shapefile if requested
if shape_overlay:
    ax.add_geometries(
        [geom], crs=ccrs.PlateCarree(),
        edgecolor="black", facecolor="none"
    )

```

Helper function to plot observation or forecast data:

```

def _plot_additional_data(tree, key, variable, init, valid_time, ax,
                          title, cmap_range):
    """Plot observation or forecast data."""
    dataset = tree[key].ds[variable]

```

```

# Handle different coordinate naming conventions
coord_key = "time" if "time" in dataset.coords else "init"
obs_init = (
    np.datetime64(valid_time.strftime("%Y-%m-%d %H:%M:%S"))
    if coord_key == "time" else init
)

# Extract data
obs_data = dataset.sel({coord_key: obs_init}).values

```

Set appropriate colormap and plot data:

```

# Set appropriate colormap
cmap = "RdBu" if key == "observation" else "Blues"

# Plot data
ax.pcolormesh(
    tree["ensemble/member_0"].ds.lon.values,
    tree["ensemble/member_0"].ds.lat.values,
    obs_data, cmap=cmap, transform=ccrs.PlateCarree(),
    vmin=cmap_range[0], vmax=cmap_range[1]
)

# Add title and formatting
ax.set_title(title, fontsize=6)
ax.set_xticks([])
ax.set_yticks([])

```

14.5 Processing Multiple Initializations

To create a complete visualization, the system processes multiple initialization dates:

```

def plot_allrows(seas5ltree, shapefile_df, params):
    """
    Create plots for all initialization dates and combine them.

    Parameters:
    seas5ltree (xarray.DataTree): Data in tree structure
    shapefile_df (GeoDataFrame): Shapefile data
    params (object): Plot parameters
    """
    # Create output directory
    output_dir = (f"{params.output_path}map_{params.region_id}_"
                  f"{params.sc_season_str}_lt{params.lead_int}")
    os.makedirs(output_dir, exist_ok=True)

    # Get region geometry
    region_geom = shapefile_df["geometry"].values[0]

    # Get all initialization dates
    inits = seas5ltree["ensemble/member_0"].ds.init.values
    plot_files = []

```

Create plots for each initialization date:

```

# Process each initialization date
for i, init in enumerate(inits):
    fig, axs, figsize = create_single_row_plot(
        seas5ltree, init, params, region_geom
    )

```

```

    )

    # Save individual plot
    output_file = f'{output_dir}/stamp_plot_{init.strftime("%Y%m%d")}.png'
    plot_files.append(output_file)
    fig.savefig(output_file, dpi=100, bbox_inches="tight")
    plt.close(fig)

# Add colorbar and title to the final figure
add_colorbar_and_title(figsize, params, output_dir)

# Merge all plots into a single image
merge_png_files(
    input_dir=output_dir,
    output_file=(f"map_{params.region_id}_{params.sc_season_str}_"
                 f"lt{params.lead_int}.pdf"),
    delete_originals=False
)

```

14.6 Merging Individual Plots

The following function combines individual timestep plots into a single visualization:

```

def merge_png_files(input_dir, output_file, delete_originals=False):
    """
    Merge individual PNG files into a single image.

    Parameters:
    input_dir (str): Directory containing PNG files
    output_file (str): Name of output file
    delete_originals (bool): Whether to delete original files
    """
    # Get all PNG files
    png_files = sorted(Path(input_dir).glob("*.png"))

    # Open all images and get dimensions
    images = []
    max_width = 0
    total_height = 0

```

Process images and create a new composite:

```

for png_file in png_files:
    with Image.open(png_file) as img:
        images.append(img.copy())
        max_width = max(max_width, img.width)
        total_height += img.height

# Create new composite image
merged_image = Image.new("RGB", (max_width, total_height), (255, 255, 255))

# Paste each image into the merged image
y_offset = 0
for img in images:
    merged_image.paste(img, ((max_width - img.width) // 2, y_offset))
    y_offset += img.height

# Save merged image
output_path = os.path.join(input_dir, output_file)
merged_image.save(output_path, dpi=(300, 300))

```

Optionally delete original files:

```
# Optionally delete original files
if delete_originals:
    for png_file in png_files:
        os.remove(png_file)
```

14.7 Example Usage

To run the map visualization process:

```
def run_map_plot(params, use_local=False):
    """
    Run the complete mapping process.

    Parameters:
    params (object): Parameters for the run
    use_local (bool): Whether to use local files

    Returns:
    str: Status message
    """
    # Get thresholds for the region and season
    threshold_dict = get_threshold(params.region_id, params.sc_season_str)

    # Prepare datasets
    obs_data, ens_data = make_obs_fct_dataset(params)
```

Process data and create plots:

```
# Calculate empirical probabilities
fct_mod, fct_sev, fct_ext = seas5l_patch_empirical_probability(
    ens_data, threshold_dict
)

# Create data tree
dstree = helper_stamp_plot(ens_data, obs_data, fct_mod, fct_sev, fct_ext)

# Get region bounds
if use_local:
    shapefile_df, extent = get_region_bounds(region_id, use_local=True)
else:
    credentials = get_credentials('coiled-data.json')
    shapefile_df, extent = get_region_bounds(
        region_id, credentials, use_local=False, buffer=0.5
    )

# Create and merge plots
plot_allrows(dstree, shapefile_df, params)
```

Create final merged PDF:

```
# Create final merged PDF
merge_png_files(
    input_dir=(f"{params.output_path}map_{params.region_id}_"
               f"{params.sc_season_str}_lt{params.lead_int}"),
    output_file=(f"map_{params.region_id}_{params.sc_season_str}_"
                 f"lt{params.lead_int}.pdf"),
    delete_originals=False
)
```



```
return "merge plot"
```

14.8 Conclusion

This approach allows for the effective visualization of both ensemble forecasts and observed data on maps with region overlays. The plots are organized by initialization date, allowing stakeholders to track forecast evolution over time and make informed decisions about anticipatory actions for drought mitigation.

Bar chart plots and Trigger visualizations

This chapter describes the implementation of bar plot visualizations for drought forecast verification as used in the anticipatory action system. The system compares SEAS5 seasonal forecasts with CHIRPS observation data to evaluate forecast skill and identify appropriate trigger values for early action.

15.1 Data Flow Overview

The bar plot visualization process follows a multi-stage workflow:

```
NetCDF Files (CHIRPS, SEAS5)
→ Verification metrics (run_xhist2d/1d)
→ CSV files with metrics
→ Bar plot generation (run_bar_plot)
→ Combined visualization with decision table
```

15.2 Components and Processing Steps

15.2.1 Data Sources and Processing

The system processes two primary data sources:

- Observation data (CHIRPS): Historical SPI values used as ground truth
- Forecast data (SEAS5): Ensemble probability forecasts for different lead times

These data are processed through verification routines that calculate metrics like hit rates, false alarm ratios, and AUROC scores for different trigger values and drought categories.

15.2.2 CSV File Structure and Organization

The verification metrics are stored in CSV files with naming convention `{region_id}_{season_str}_{lead_time}_{category}.csv` and combined into `{region_id}_{season_str}_{lead_time}.csv` for each lead time.

Each combined CSV contains important columns:

- `trigger_value`: Probability threshold for triggering an alert
- `cat`: Drought category (mod, sev, ext - corresponding to mild, moderate, severe)
- `td`: Trigger decision flag (1 = selected as trigger)
- `hit_rate`: Percentage of drought events correctly forecasted

- `false_alarm_ratio`: Percentage of false alarms
- `hits`: Number of correctly forecasted events
- `misses`: Number of missed events
- `FA`: Number of false alarms
- `CN`: Number of correct negatives

Example CSV content (simplified):

```
trigger_value,cat,td, hit_rate, false_alarm_ratio, hits, misses, FA, CN
20.2,mod,1, 73.3, 16.7, 11, 4, 2, 25
35.4,mild,1, 66.7, 8.7, 10, 5, 3, 24
15.2,sev,1, 75.0, 9.8, 6, 2, 2, 32
```

15.2.3 Bar Plot Generation Process

The `run_bar_plot` function in `utils_plots.py` orchestrates the bar plot generation. This function:

1. Retrieves thresholds for the region and season
2. Gets observation and forecast data
3. Processes lead times 2, 3, and 4 separately
4. Creates visualizations for each lead time and observation data
5. Combines all plots into a final visualization

15.2.4 Key Subroutines

Several functions work together to create the final visualization:

- `run_bar_plot_df()`: Retrieves and processes observation and forecast data
- `generate_trigger_dict()`: Extracts trigger values from verification CSV files
- `plot_obs_chart_with_triggers()`: Creates Altair visualizations with threshold lines
- `bar_stitch_plot()`: Combines individual plots into a comprehensive visualization

15.2.5 Visualizing Different Lead Times

Each lead time's data is processed separately and presented in a structured layout:

```
+-----+-----+
| Observations | Lead Time 2 |
+-----+-----+
| Decision     | Lead Time 3 |
| Table       |              |
+-----+-----+
| Empty Space  | Lead Time 4 |
+-----+-----+
```

The decision table combines information from all lead times, showing:

- Selected trigger values for each drought category
- Performance metrics for each trigger
- Lead time information

15.3 Visual Elements and Interpretation

The visualization includes several key elements:

- Observation plot: Shows historical SPI values with positive values in blue and negative in red
- Forecast probability plots: Show ensemble probability forecasts with bars colored by drought category
- Threshold lines: Horizontal lines indicating the selected trigger values
- Decision table: Displays trigger values with their performance metrics across lead times

15.4 Technical Implementation Details

15.4.1 Trigger Selection and Extraction

The `generate_trigger_dict()` function reads CSV files to extract chosen triggers. This function:

1. Reads the CSV file for the current lead time
2. Filters for selected triggers (where `td == 1`)
3. Creates a trigger dictionary mapping drought levels to probability values
4. Formats a decision dataframe for display
5. Returns both the trigger dictionary and decision dataframe

15.4.2 Plot Creation with Altair

The system uses Altair to create interactive, layered visualizations through the `plot_obs_chart_with_triggers()` function, which:

1. Creates different base charts for observations versus forecasts
2. Adds threshold lines for each trigger value
3. Combines the elements into layered visualizations
4. Returns the final chart object

15.4.3 Final Visualization Assembly

The `bar_stitch_plot()` function combines all visualization components:

1. Extracts individual plot components
2. Creates a decision table from combined dataframes
3. Assembles the final layout with observations, forecasts, and decision table
4. Saves the visualization as a PNG file

15.5 Practical Applications and Benefits

The generated bar plot visualization serves several important functions:

- Historical comparison: Shows how forecasts would have performed in past years

- Lead time evaluation: Allows comparison of forecast skill at different lead times
- Trigger assessment: Visualizes the performance of selected trigger values
- Decision support: Provides a comprehensive view for stakeholder decision-making

By combining observation data, forecast probabilities, and performance metrics in a single visualization, the system enables informed decisions about which trigger values to use for anticipatory action against droughts.

15.6 Conclusion

The bar plot visualization system is a critical component of the drought anticipatory action framework. It translates complex verification metrics into an intuitive visual format that facilitates stakeholder understanding and decision-making. By clearly showing when drought alerts would have been triggered historically and how accurate those alerts would have been, it builds confidence in the trigger system and supports effective early action to mitigate drought impacts.

